# Blood Supply of the CNS

**Vincy Mathew**, OMS-IV
Academic Medicine Scholar 2024
vmathe03@nyit.edu

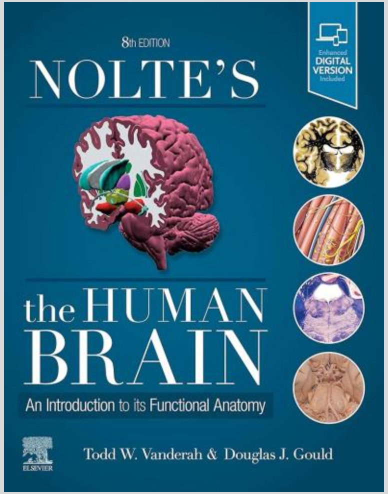
**Jennifer Xie, Ph.D.**
Department of Biomedical and Anatomical Sciences
Jennifer.Xie@nyit.edu

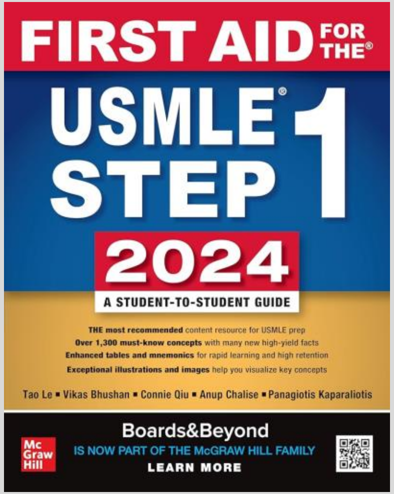
Do.
Make.
Heal.
Innovate.
Reinvent the Future.

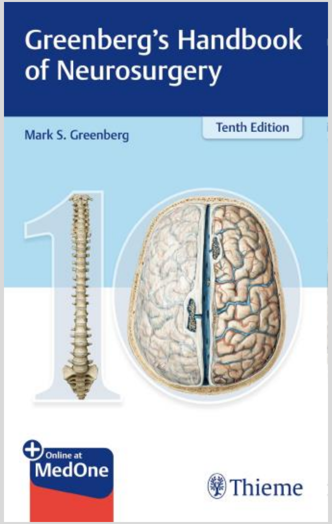
# References

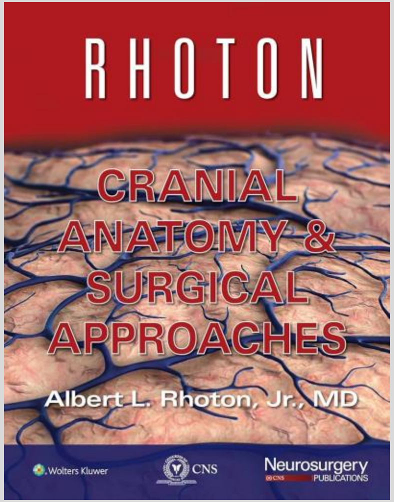
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine



# Session Objectives

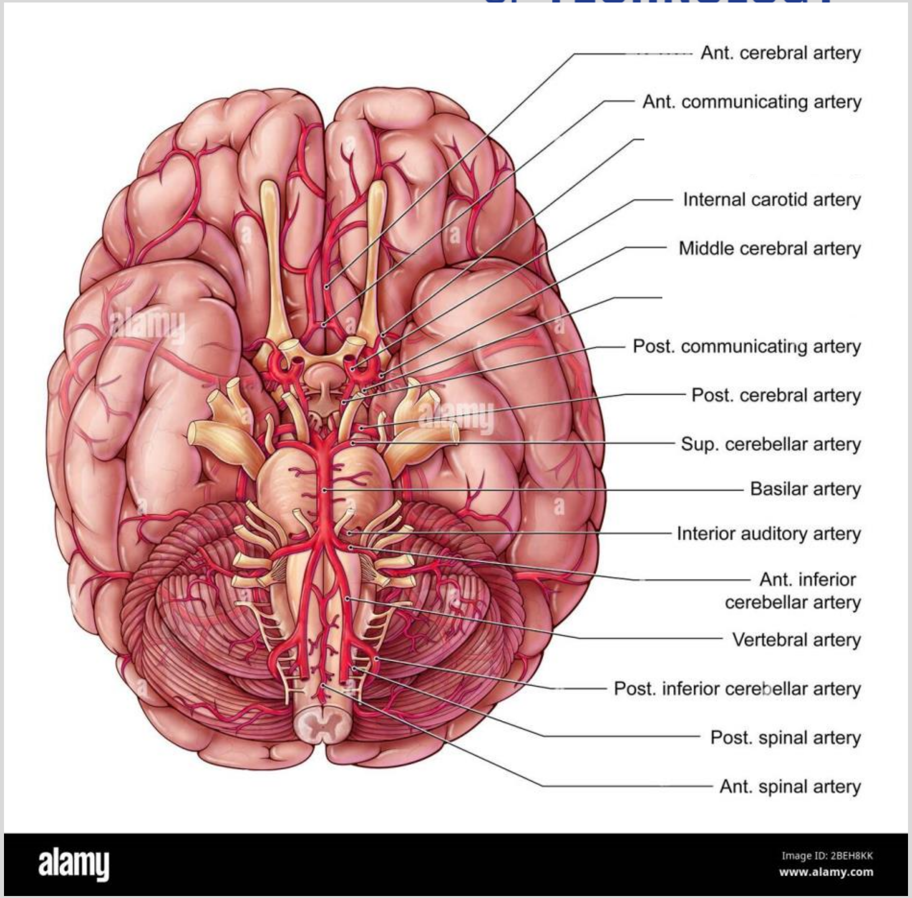
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

1. Identify the four major vessels that supply blood to the brain.
2. Differentiate between the anterior and posterior circulations of the brain, including their components.
3. Describe the formation and function of the Circle of Willis.
4. Locate and identify the regions of the brain supplied by each major artery using gross anatomical landmarks.
5. Explain the structures of the cerebral venous system and their roles in venous drainage.
6. Classify and differentiate the types of intracranial hemorrhages based on their origin of bleed and the compartments of the brain they affect.

# Arteries you should know by the end of this lecture

- CCA, common carotid artery
- ECA, external carotid artery
- ICA, internal carotid artery
- MCA, middle cerebral artery
- ACA, anterior cerebral artery
- ACom, anterior communicating artery
- PCom, posterior communicating artery
- VA, vertebral artery
- BA, basilar artery
- PCA, posterior cerebral artery
- AICA, anterior inferior cerebellar artery
- PICA, posterior inferior cerebellar artery
- SCA, superior cerebellar artery
- PA, pontine artery
- ASA, anterior spinal artery
- PSA, posterior spinal artery



# Arteries you should know by the end of this lecture

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

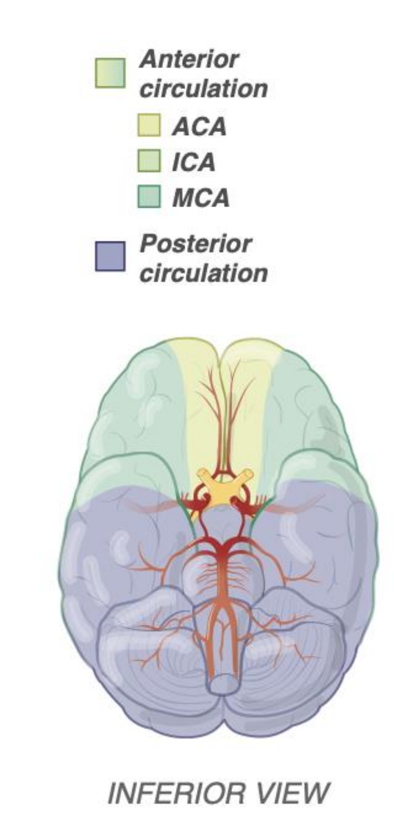
- CCA, common carotid artery
- ECA, external carotid artery
- ICA, internal carotid artery
- MCA, middle cerebral artery
- ACA, anterior cerebral artery
- ACom, anterior communicating artery
- PCom, posterior communicating artery
- VA, vertebral artery
- BA, basilar artery
- PCA, posterior cerebral artery
- AICA, anterior inferior cerebellar artery
- PICA, posterior inferior cerebellar artery
- SCA, superior cerebellar artery
- PA, pontine artery
- ASA, anterior spinal artery
- PSA, posterior spinal artery

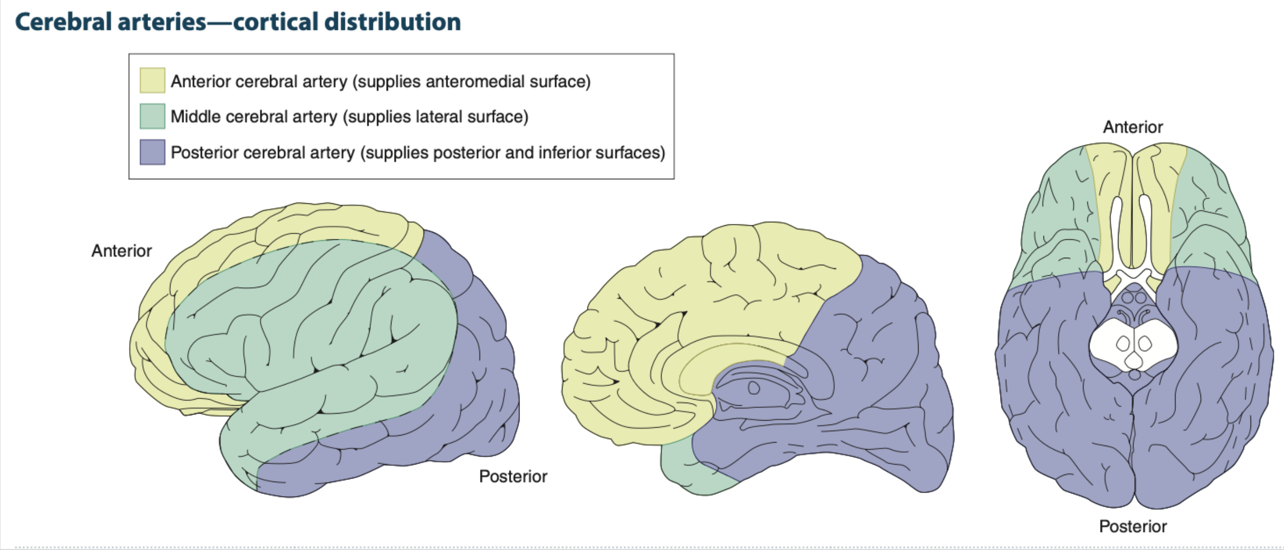
1. **Parent vessel**→ where does it come from?
2. **Destination** → is it giving another branch or supplying a brain area?
3. **Adjacent structures** → what other neurovascular structures are relative to it?
4. **Blood supply** →what part of the brain does it supply blood to?

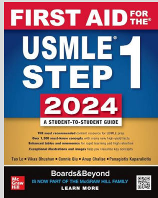
# Master this by the end

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

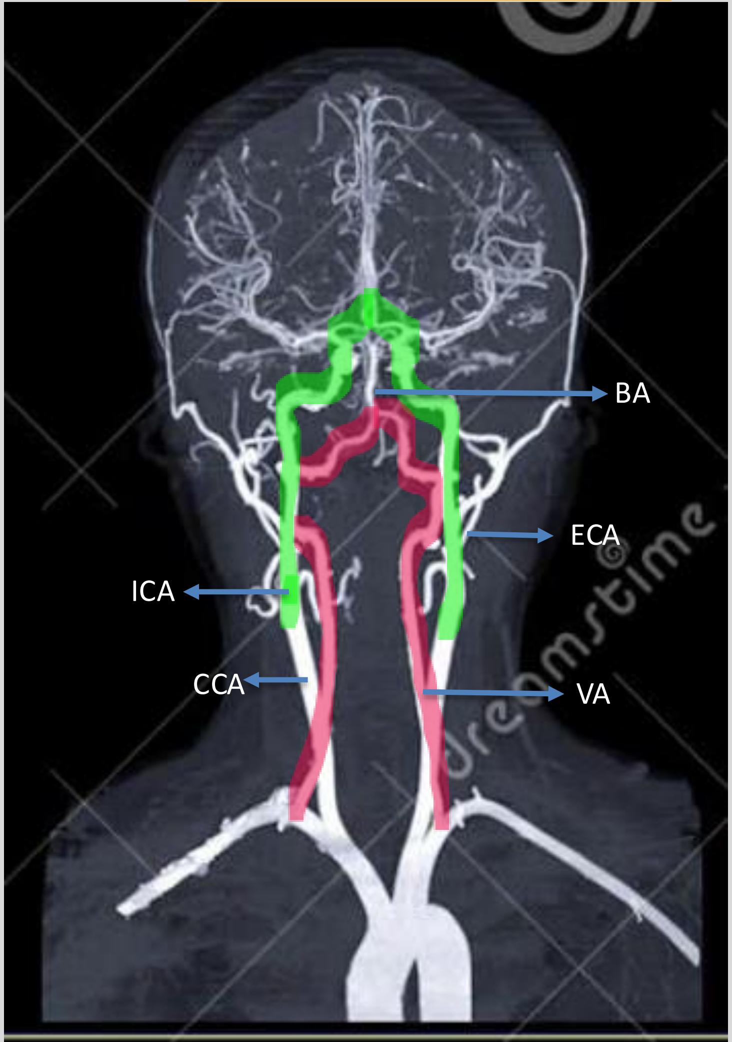


# The four major arteries

- **2 Internal carotid arteries** (**ICA**)→ anterior circulation
  - ~80% of blood supply
  - Telencephalon and diencephalon
- **2 Vertebral arteries** (**VA**) → posterior circulation
  - ~20%
  - Diencephalon
  - Brainstem (midbrain, pons, medulla), cerebellum
- Anterior and posterior circulations connected by **posterior communicating arteries (PCom)**



# The four major arteries

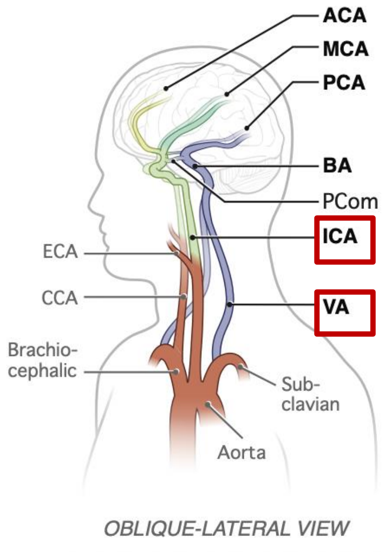
**2 Internal carotid arteries** (**ICA**)→ anterior circulation

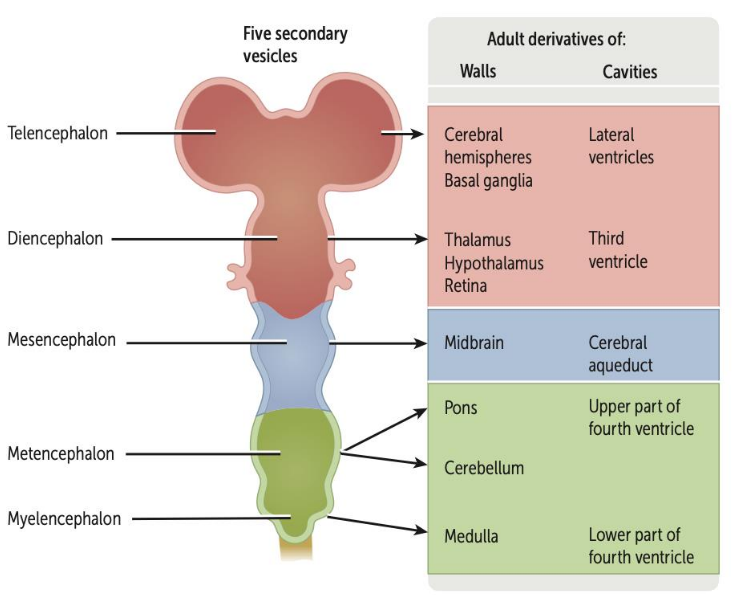
Telencephalon and diencephalon

**2 Vertebral arteries** (**VA**) → posterior circulation

Diencephalon

Brainstem (midbrain, pons, medulla), cerebellum

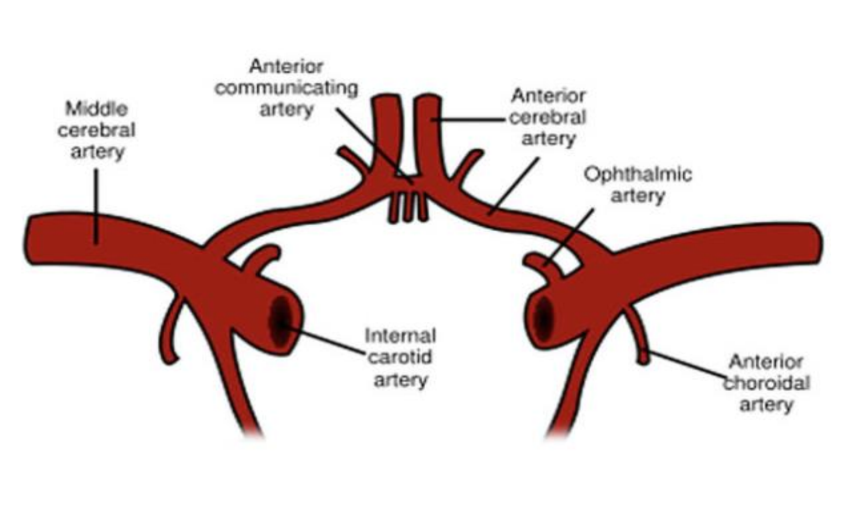


NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

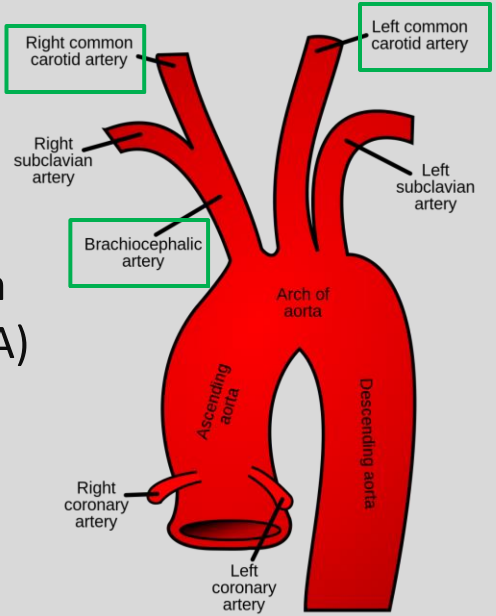
# ANTERIOR CIRCULATION

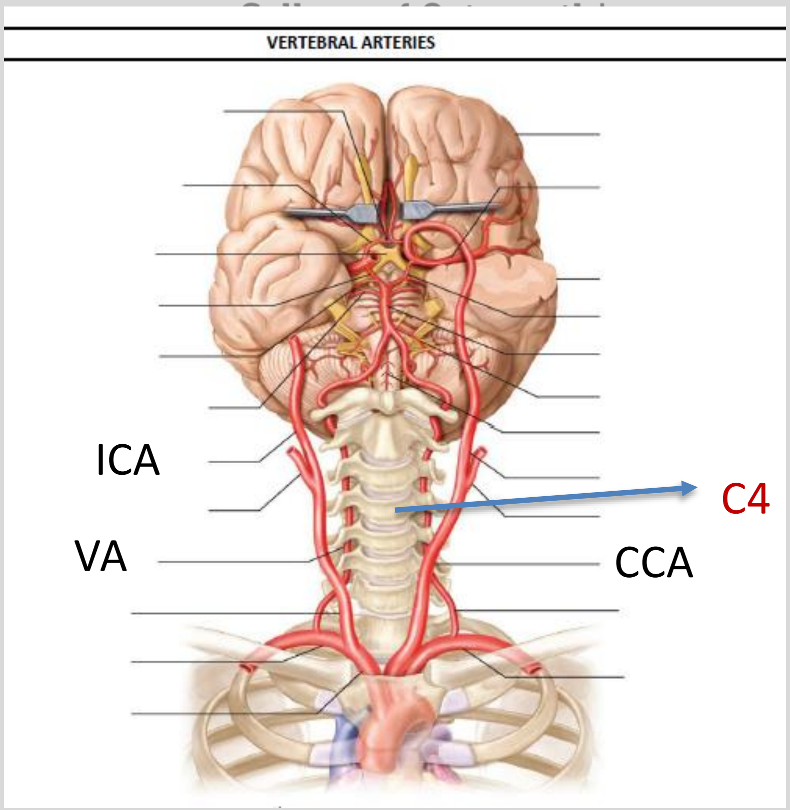


# Internal Carotid Artery (ICA)

NEW YORK INSTITUTE OF TECHNOLOGY

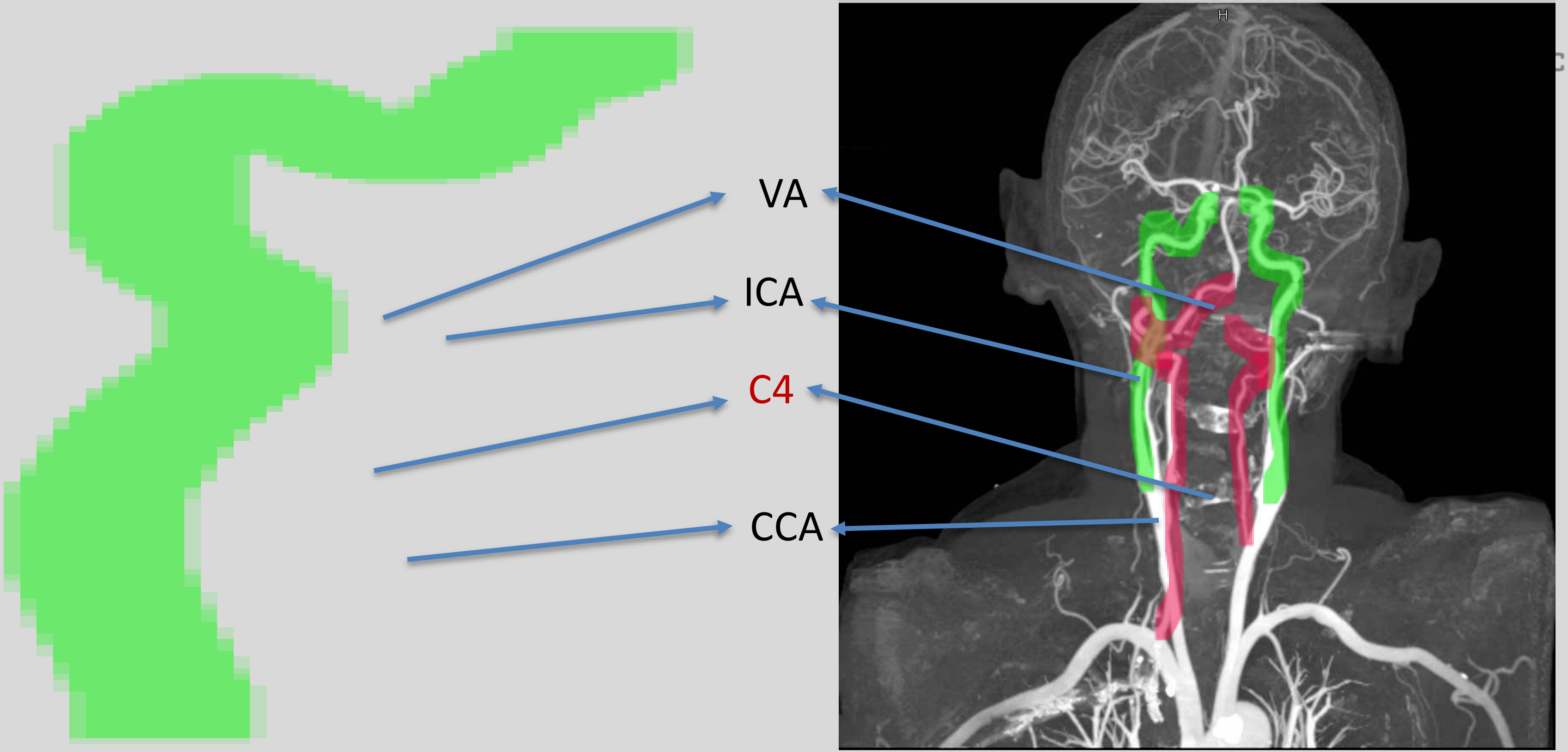
- Arch of the aorta→
  - brachiocephalic trunk → right common carotid artery (CCA)
  - Left right common carotid artery (CCA)
- CCA bifurcates (bifourcates) into external carotid artery (ECA) and internal carotid artery (ICA) at the level of C4 vertebra.



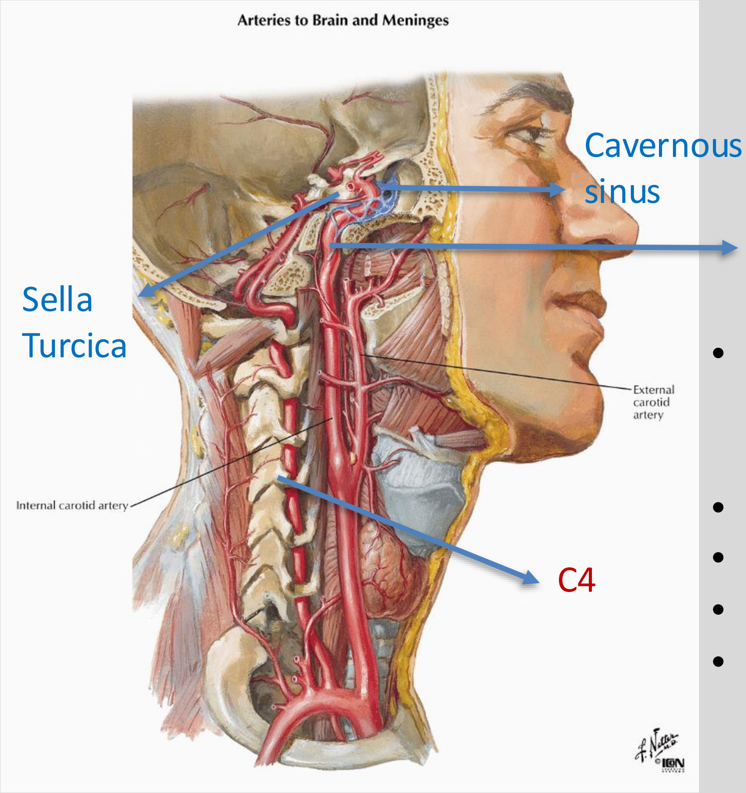



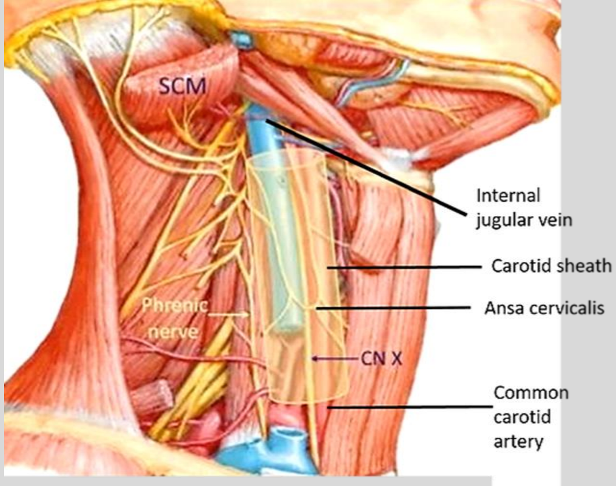
# Internal Carotid Artery (ICA)

NEW YORK INSTITUTE OF TECHNOLOGY



# Internal Carotid Artery (ICA)




- ICA Travels in the **carotid sheath** in the neck and enters the skull through the '**carotid canal**' in the petrous part of temporal bone.
- Once inside the skull→ **intracranial ICA**
- It travels through the **cavernous sinus** next to Sella turcica.
- Once it traverses the dura → **intradural ICA**
- Intradural ICA branches: (**OPAM**)
    1. Ophthalmic artery
    2. Posterior Communicating artery
    3. Anterior Cerebral artery
    4. Middle Cerebral artery

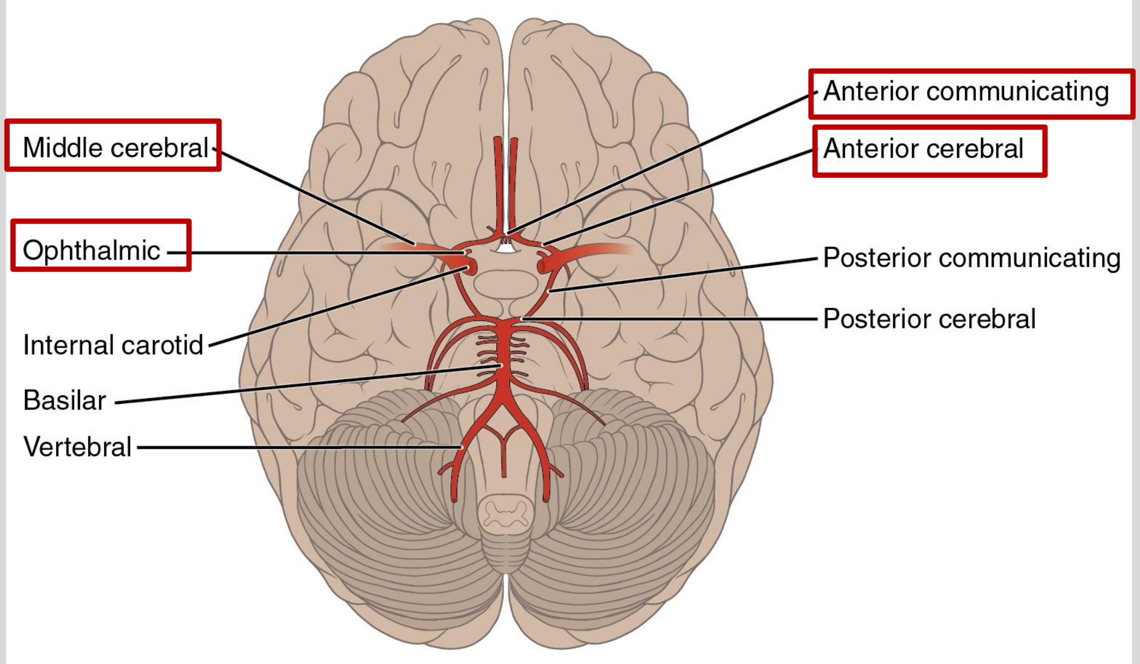
# Intradural ICA Branches

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

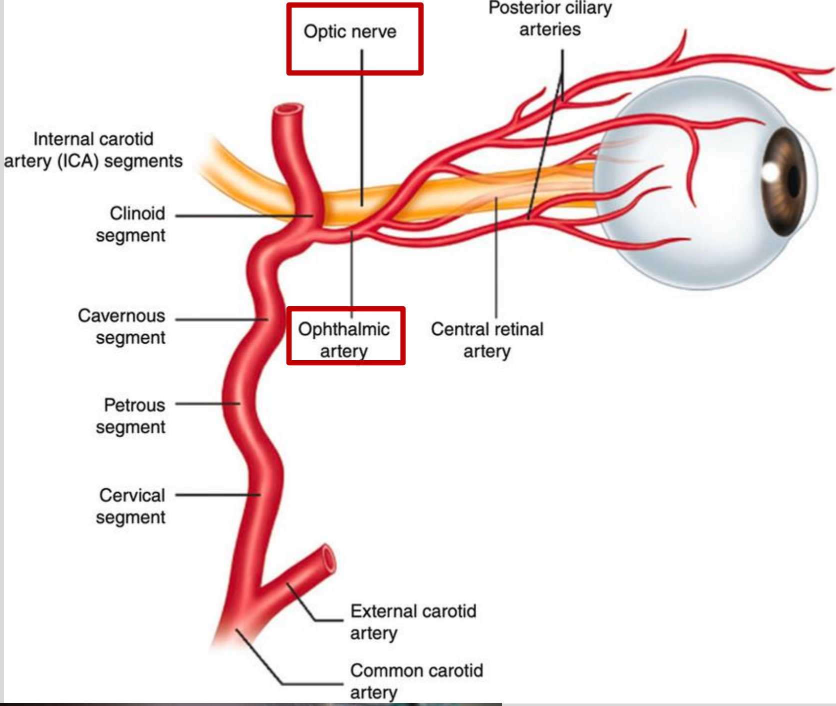
**OPAM:**

1. **O**phthalmic Artery
2. **P**osterior Communicating Artery
3. **A**nterior Cerebral Artery
4. **M**iddle Cerebral Artery

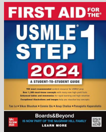


# **OPAM:** Ophthalmic Artery

- **1$^{ST}$ branch** of intradural ICA
- **Supplies the eye** and surrounding structures
- Travels with ophthalmic vein and **optic nerve**
- **Occlusion of ophthalmic artery**: can be the first place a dislodged clot traveling in the ICA get lodged: can cause **Amaurosis fugax**
  - Transient painless vision loss due to lack of perfusion on ophthalmic a.
  - "a curtain coming down over my eye"
  - Can be an early indicator of a stroke

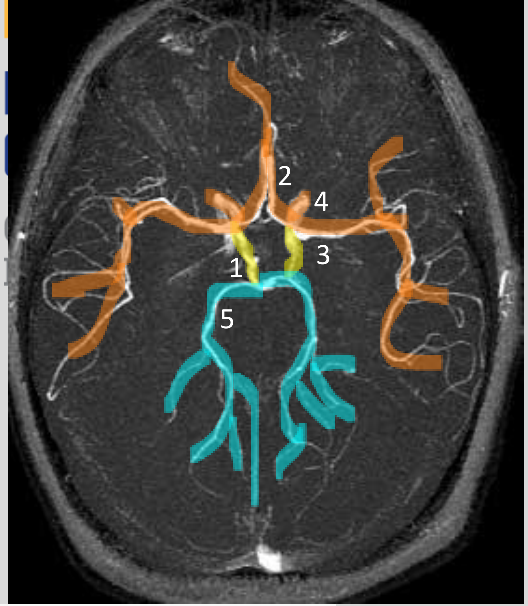


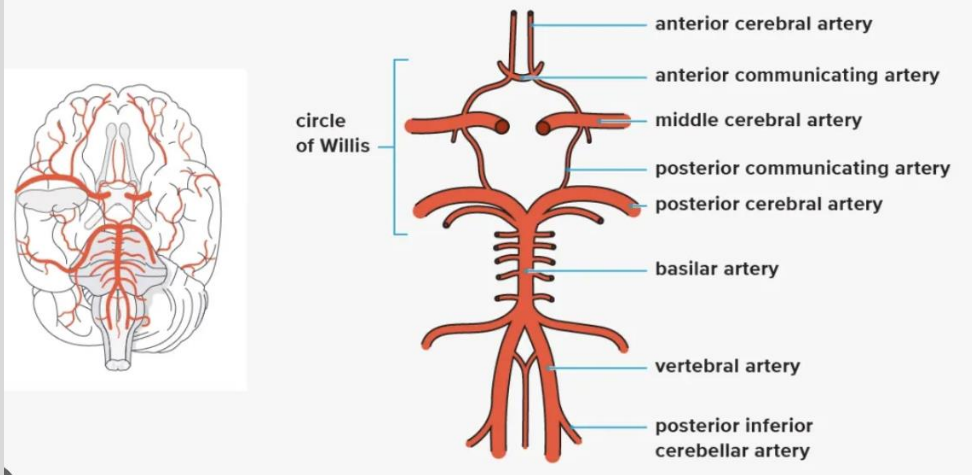

More on this in the cerebrovascular pathology lecture later..



# OPAM: Posterior Communicating Artery (PCom)

- As the name suggests, PCom **communicates** the anterior and posterior circulation that forms the circle of Willis.
- Patency of PCom is crucial for adequate circulation of the circle of Willis.

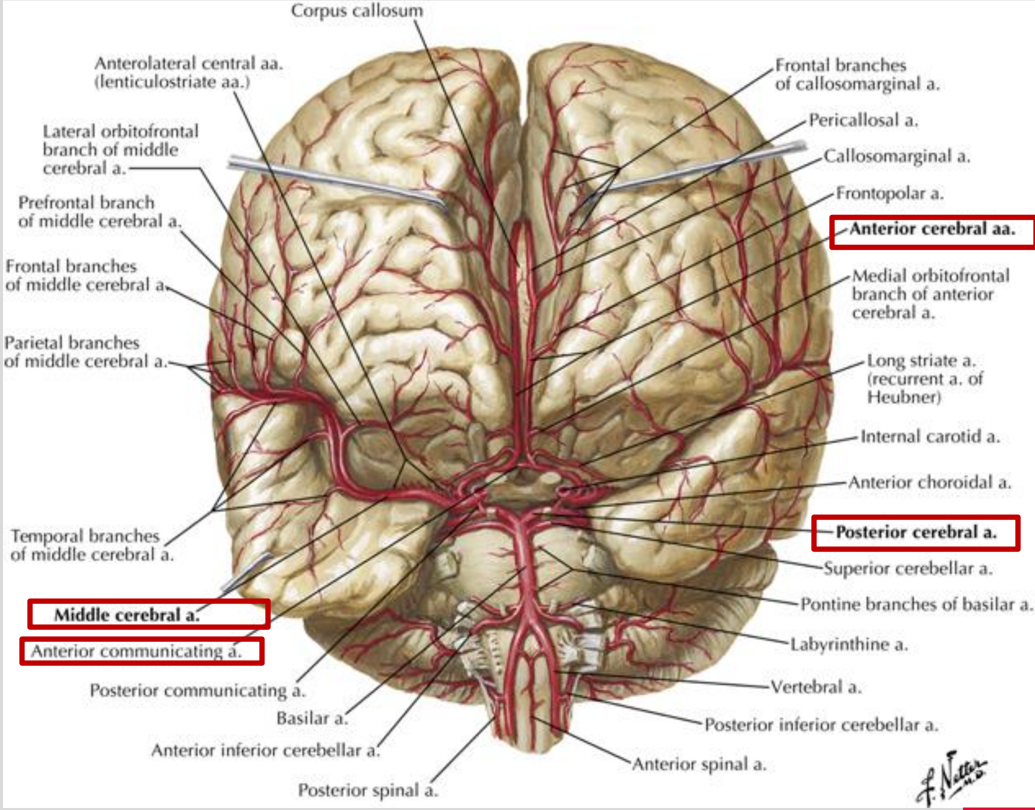


# OPAM: Anterior Cerebral Artery (ACA)

NEW YORK INSTITUTE OF TECHNOLOGY

- **ICA bifurcates** into **anterior cerebral artery (ACA) medially** and **middle cerebral (MCA) artery laterally**
- Bifurcation happens under the frontal lobe, in the medial border of the sylvian fissure.
- The left and right ACA communicate with each other through the **anterior communicating artery (ACom)**.
- The ACA then travels in the **interhemispheric** or the longitudinal fissure to **supply blood to the medial and anterior portions of the frontal, temporal and parietal lobes of the cerebrum.**

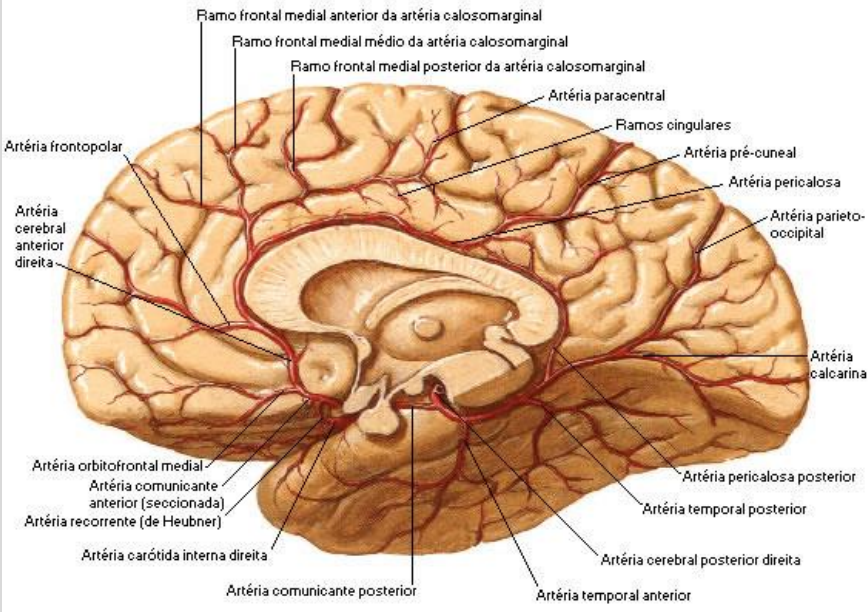


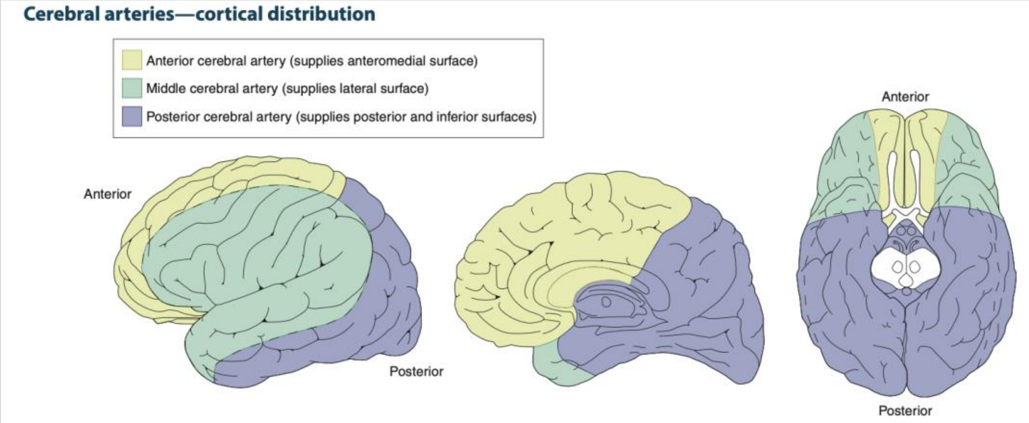
# OPAM: Anterior Cerebral Artery (ACA)

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Travels antero-medially above **the optic nerve** and **chiasm.**
- **Supply blood to the medial and anterior portions of the frontal and parietal lobes of the cerebrum.**

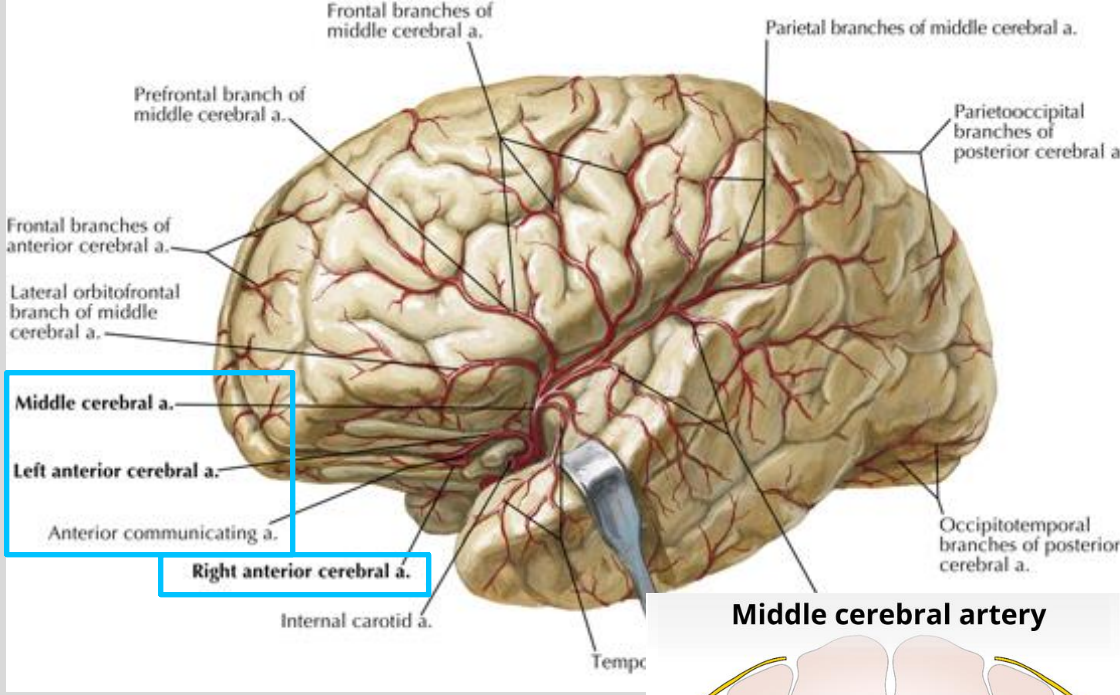


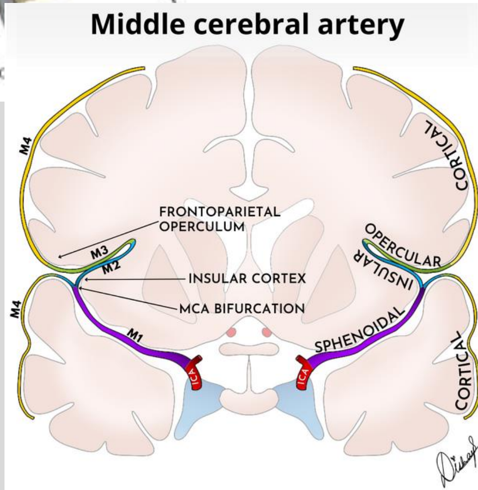




FIRST AID FOR THE USMLE STEP 1 2024

# OPAM: Middle Cerebral Artery (MCA)

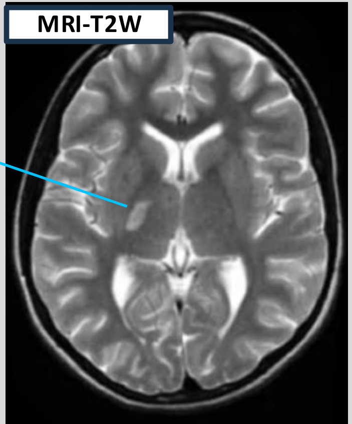
- **ICA bifurcates** into **anterior cerebral artery (ACA) medially** and **middle cerebral (MCA) artery laterally**
- The middle cerebral artery is **the larger terminal branch** of the internal carotid artery.
- Bifurcations happens on the medial border of the sylvian fissure.
- The MCA then follows the borders of the **sylvian fissure laterally**, **wraps around the insular cortex** and emerge on the cortical surface of the brain **supplying much of the lateral parts of the frontal, temporal and parietal lobe.**

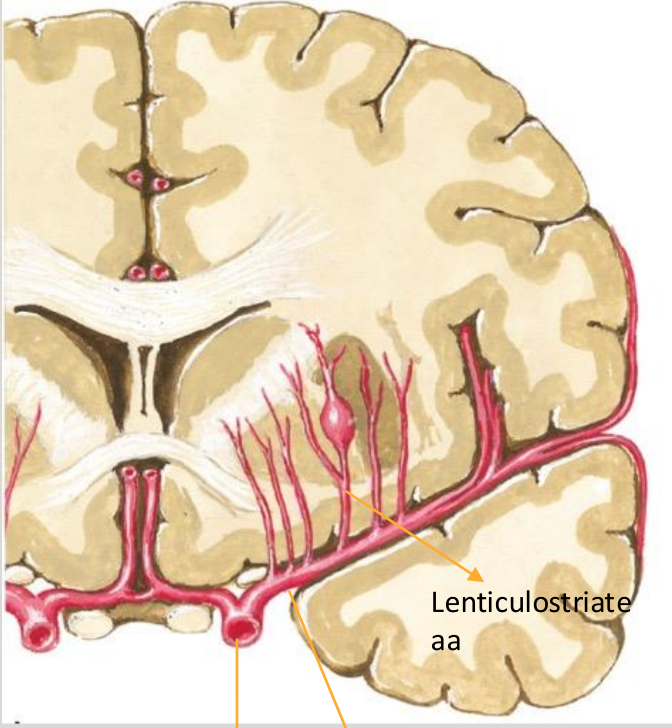


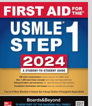



# OPAM: Middle Cerebral Artery (MCA)

- An important deep branch of MCA is the tiny **lenticulostriate arteries** that supply the **basal ganglia**.
- These arteries are vulnerable to the formation of small aneurysms, especially in the setting of **chronic, uncontrolled hypertension**. When these aneurysms rupture, they can lead to hemorrhages known as **Charcot-Bouchard aneurysm.**
- Lenticulostriate arteries are also **susceptible to ischemic strokes** given their size and are know as **lacunar strokes** , named for the characteristic "**lake-like**" appearance of the resulting cavities in the brain tissue on MRI of the brain.

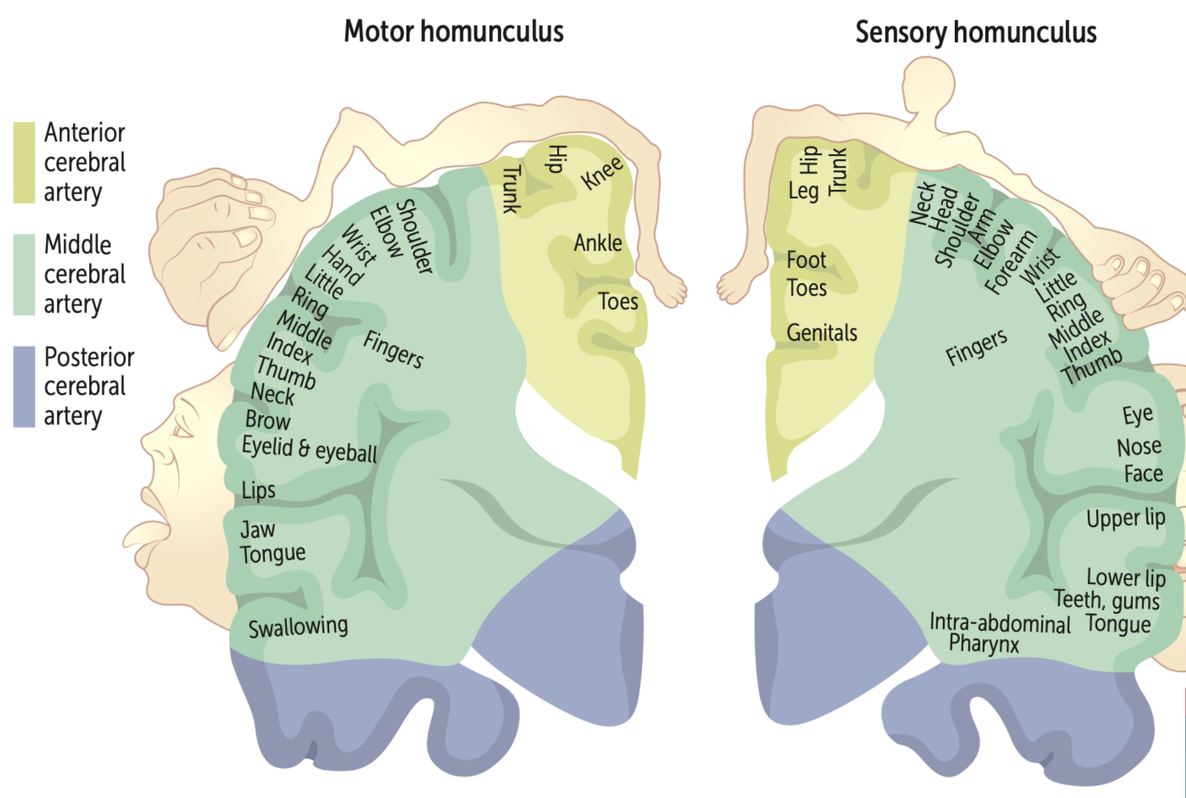




More on this in the cerebrovascular pathology lecture later..

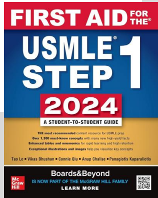
# Motor and Sensory Homunculus-
# MCA & ACA

- ACA feed motor and sensory areas for: lower extremity and trunk
- MCA feed motor and sensory areas for: upper extremity and face
- How do you expect a patient with stroke of either artery to present?




More on this in the cerebrovascular pathology lecture later..

NEW YORK INSTITUTE OF TECHNOLOGY

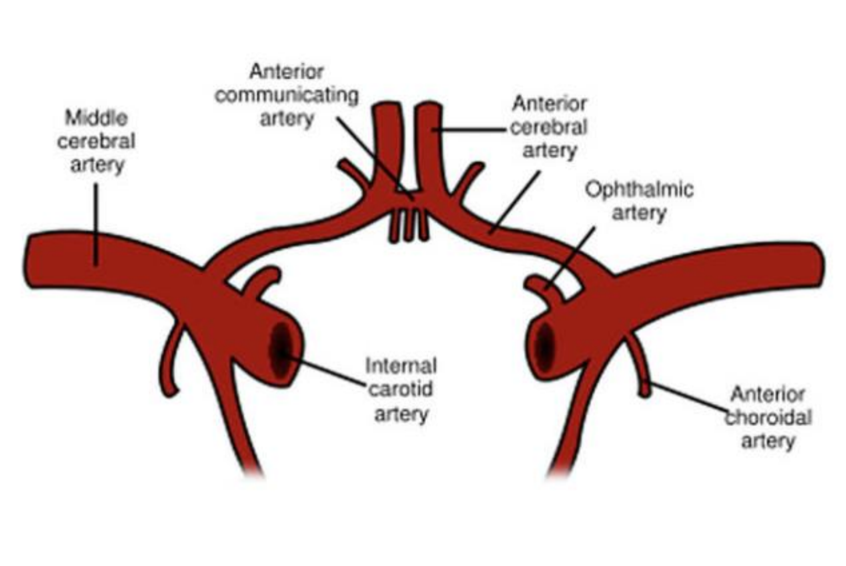


# Knowledge check:

# ANTERIOR CIRCULATION

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

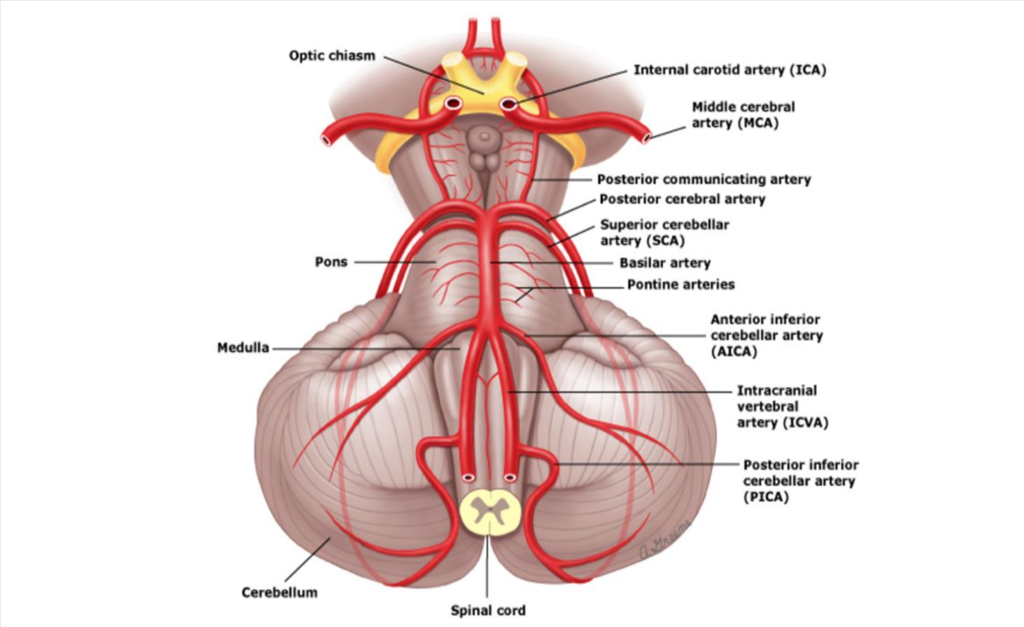



1. **Parent vessel**→ where does it come from?
2. **Destination** → is it giving another branch or supplying a brain area?
3. **Adjacent structures** → what other neurovascular structures are relative to it?
4. **Blood supply** →what part of the brain does it supply blood to?

NEW YORK INSTITUTE OF TECHNOLOGY

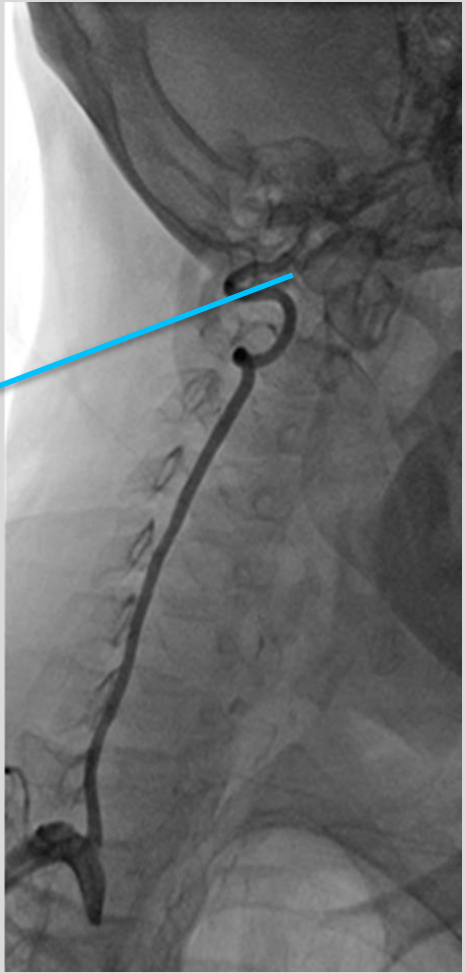
College of Osteopathic Medicine

# POSTERIOR CIRCULATION



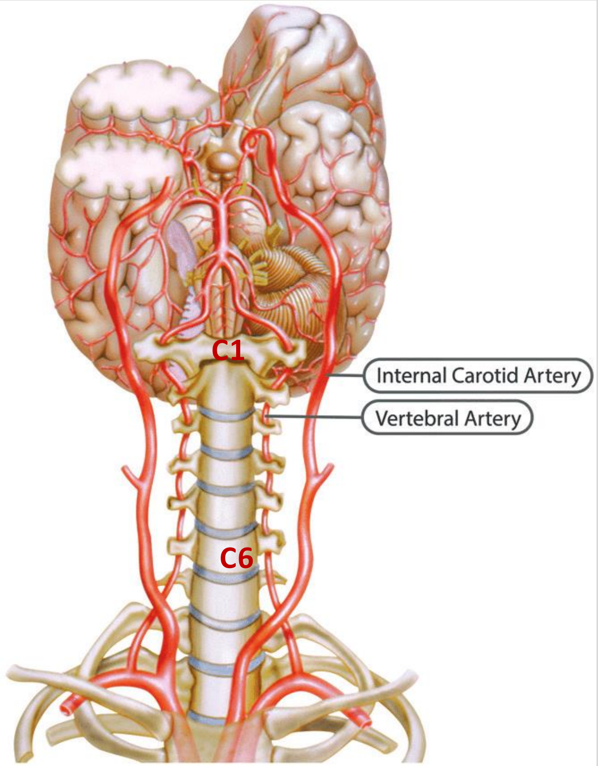
# Vertebral Artery (VA)

- **Subclavian artery → vertebral artery** (VA)
- Travel through the **transverse foramen of C6-C1 vertebrae**
- **Enter the cranium** via the **foramen magnum, lateral and ventral to the brainstem.**
- The **left and right vertebral arteries fuse** to become the **basilar artery (BA)** below the level of pons.



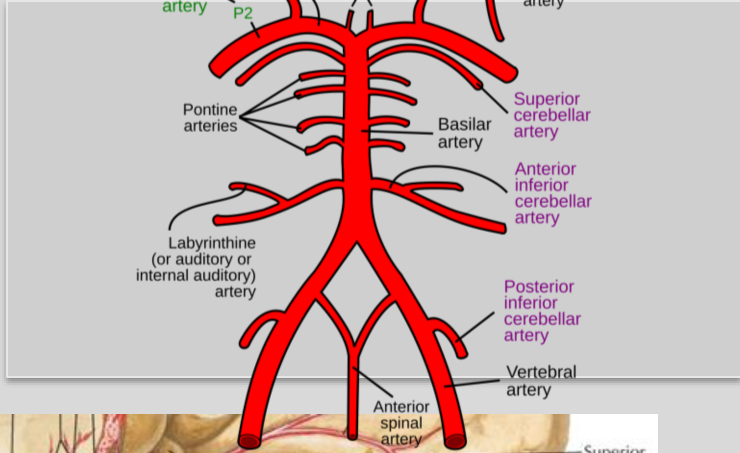
NEW YORK INSTITUTE OF TECHNOLOGY

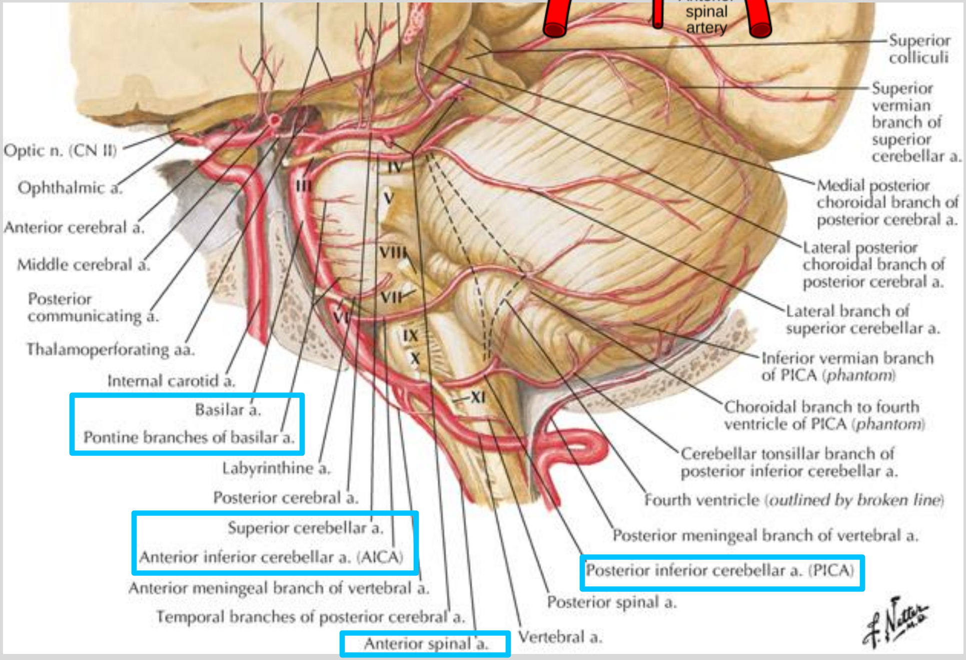
College of Osteopathic Medicine



# Vertebrobasilar system

- The vertebral arteries and its branches + basilar arteries and its branches → **vertebrobasilar system**: makes up the infratentorial portion of the posterior circulation.
- Vertebral artery branches:
  - Posterior inferior cerebellar artery (PICA)
    - Supply: inferior and medial cerebellum
  - Anterior spinal artery (ASA)
    - Supply: anterior spinal cord
- Basilar artery branches:
  - Anterior inferior cerebellar arteries (AICA)
  - Supply: anterior and lateral cerebellum
  - Pontine arteries (PA)
    - Supply: basilar pone
  - Superior cerebellar arteries (SCA)
    - Supply: superior- medial and superolateral cerebellum



# New angiographic variant of PICA – Mathew et al.

- The cerebellar arteries are among the most anatomical variable intracranial arteries.
- New, previously unreported angiographic variant of PICA.
- Right PICA originated directly from the right SCA.
- Left PICA originated from the basilar trunk, distal to the origin of the left AICA.
- CEREBROVASCUALR SYSTEM IS WONDERFULLY DIVERSE!

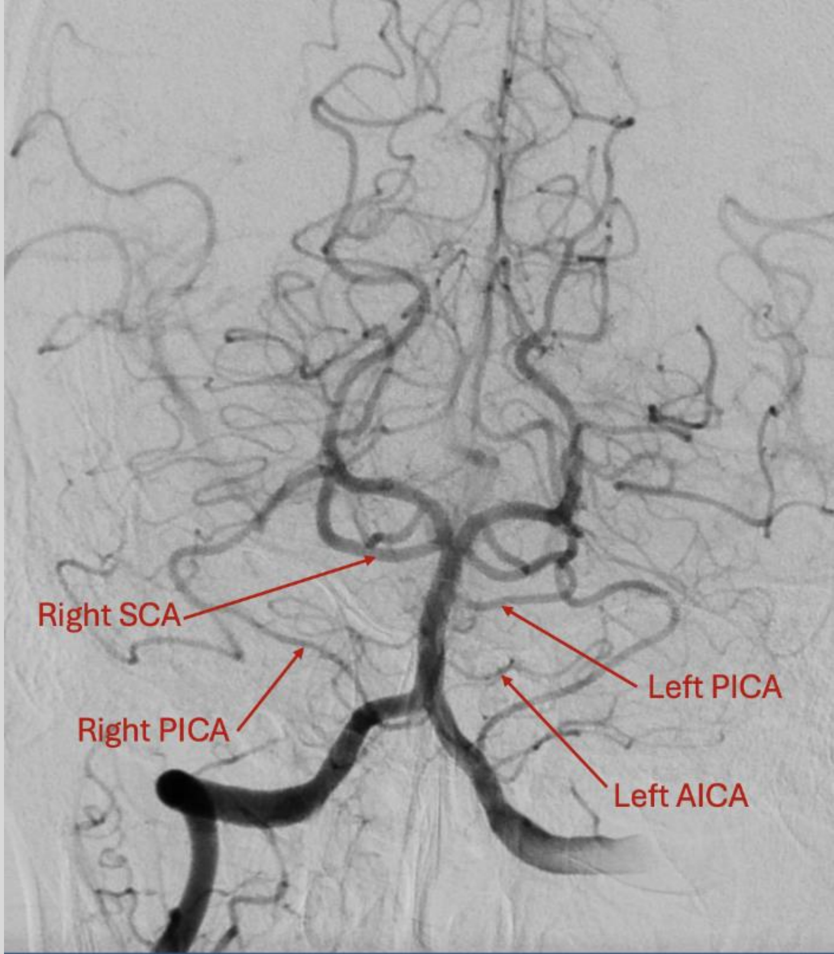


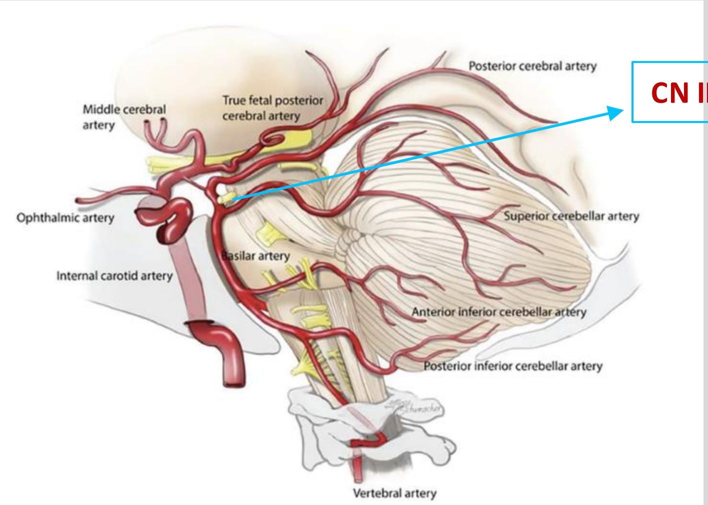

**Figure 1. Vertebral artery angiography shows the right PICA originating from the right SCA, and the left PICA arising directly from the basilar trunk, distal to the origin of the left AICA.**

# Posterior Cerebral Artery (PCA)

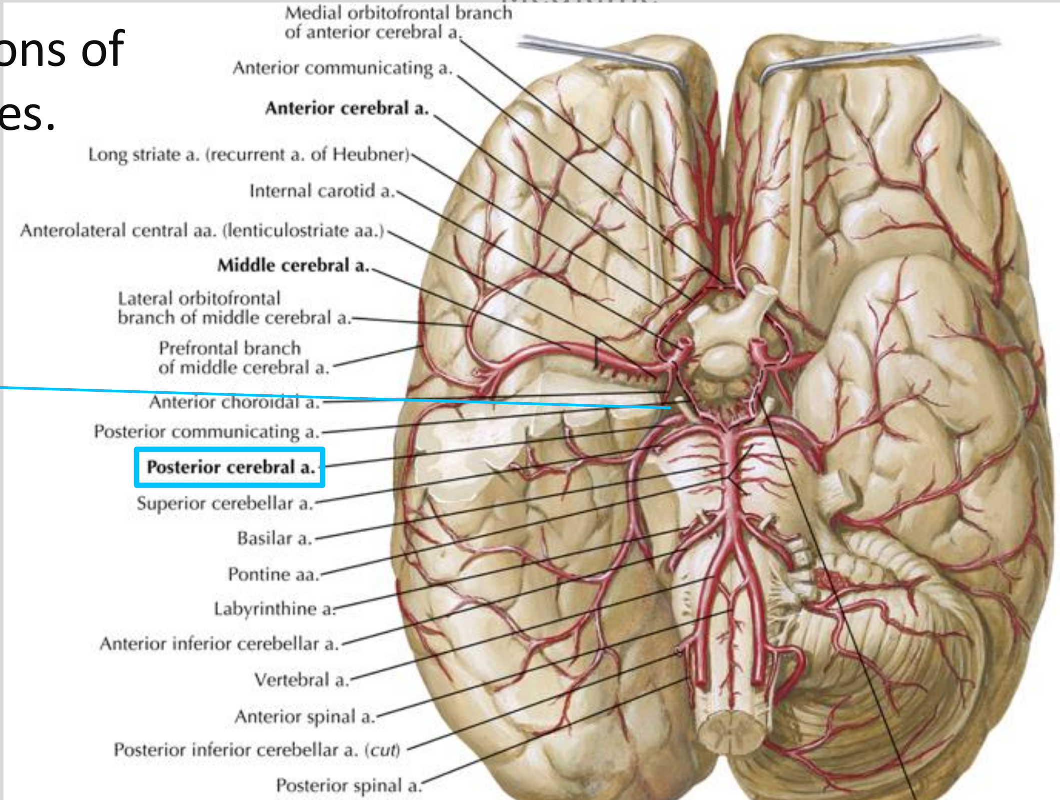
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- **Posterior cerebral artery (PCA)** is the **terminal branch** of **basilar artery**.
- **Supplies**: inferior and medial portions of the cerebrum and the occipital lobes. **Above the tentorium**
- **CN III travels b/w PCA and SCA**

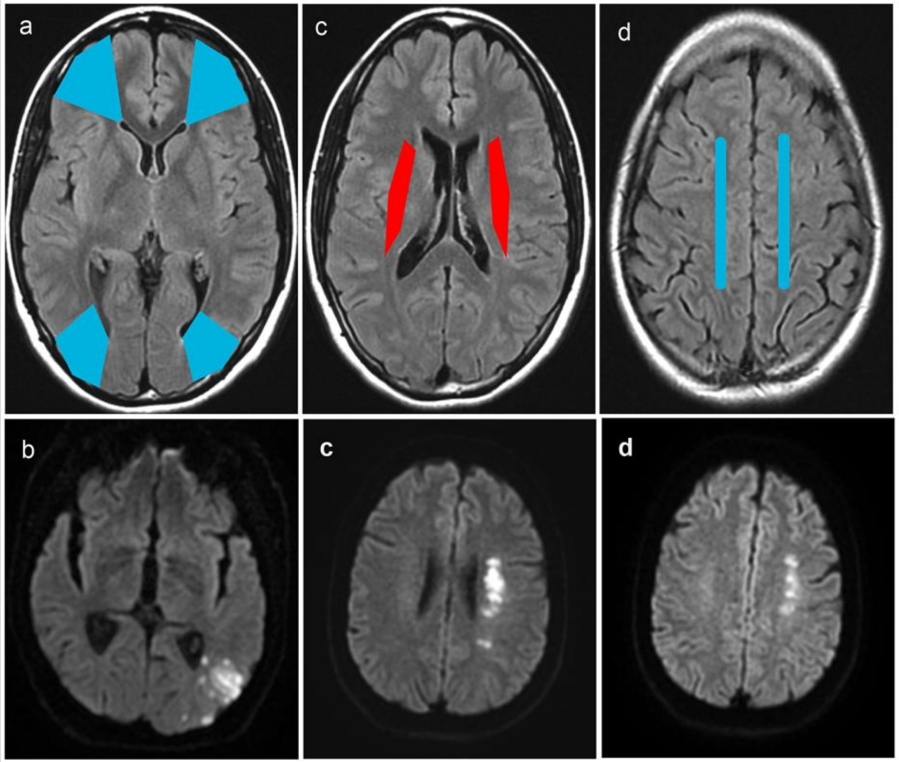



CN III

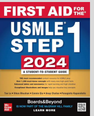


# Watershed Area

- Regions in the brain situated at the **border zones** between major cerebral arteries, particularly the anterior, middle, and posterior cerebral arteries.
- They **are prone to infarction** during **global hypoperfusion**.
- Cortical Watershed Zones:
    - Outer cortical areas where **ACA, MCA, and PCA territories meet.**
- Subcortical Watershed Zones:
    - Deep brain regions, typically between the deep vascular territories of the MCA and ACA



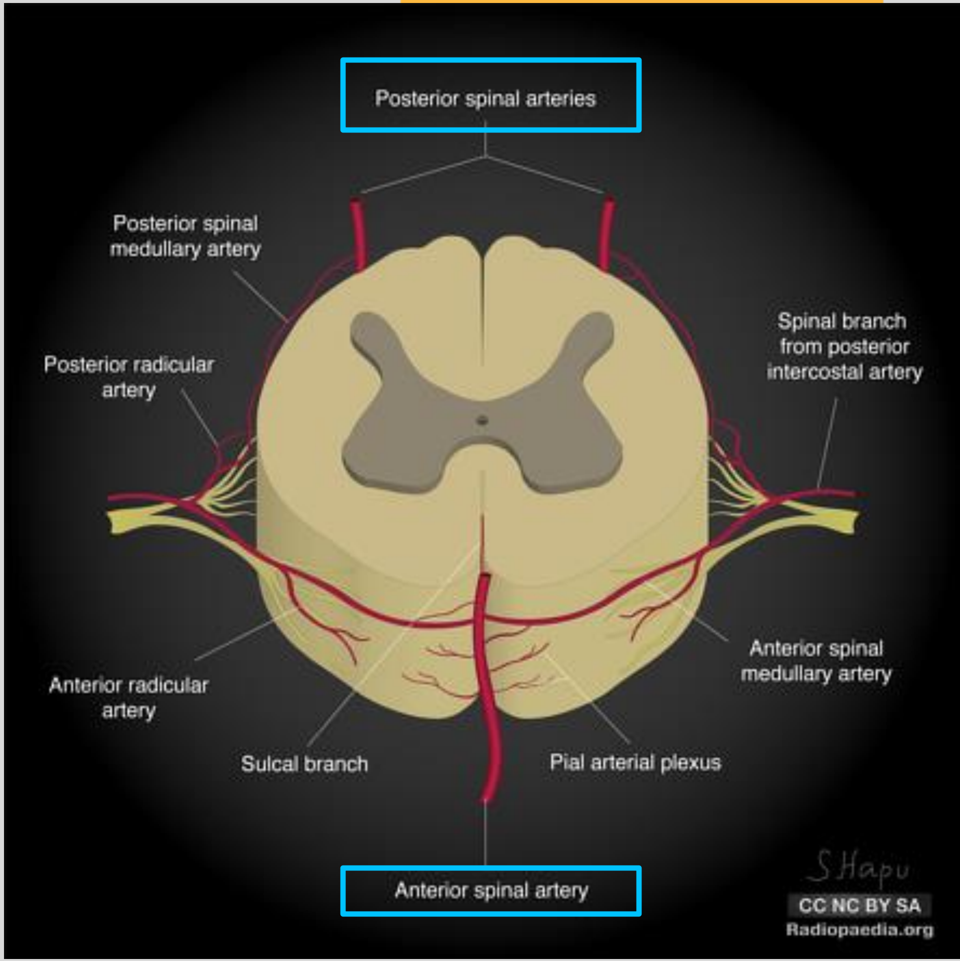

Watershed territories, on MRI (FLAIR); in blue: cortical watershed, in red: internal watershed; (a) frontal WSI: between the anterior and the middle cerebral arterial territories, (a, b) occipital WSI: at the boundaries of the middle and the posterior arterial territories; (c) internal WSI: between territories of superficial and deep perforators of the middle cerebral artery; (d) paramedian WSI: between the anterior and the middle cerebral arterial territories.




More on this in the cerebrovascular pathology lecture later..

# Spinal arteries

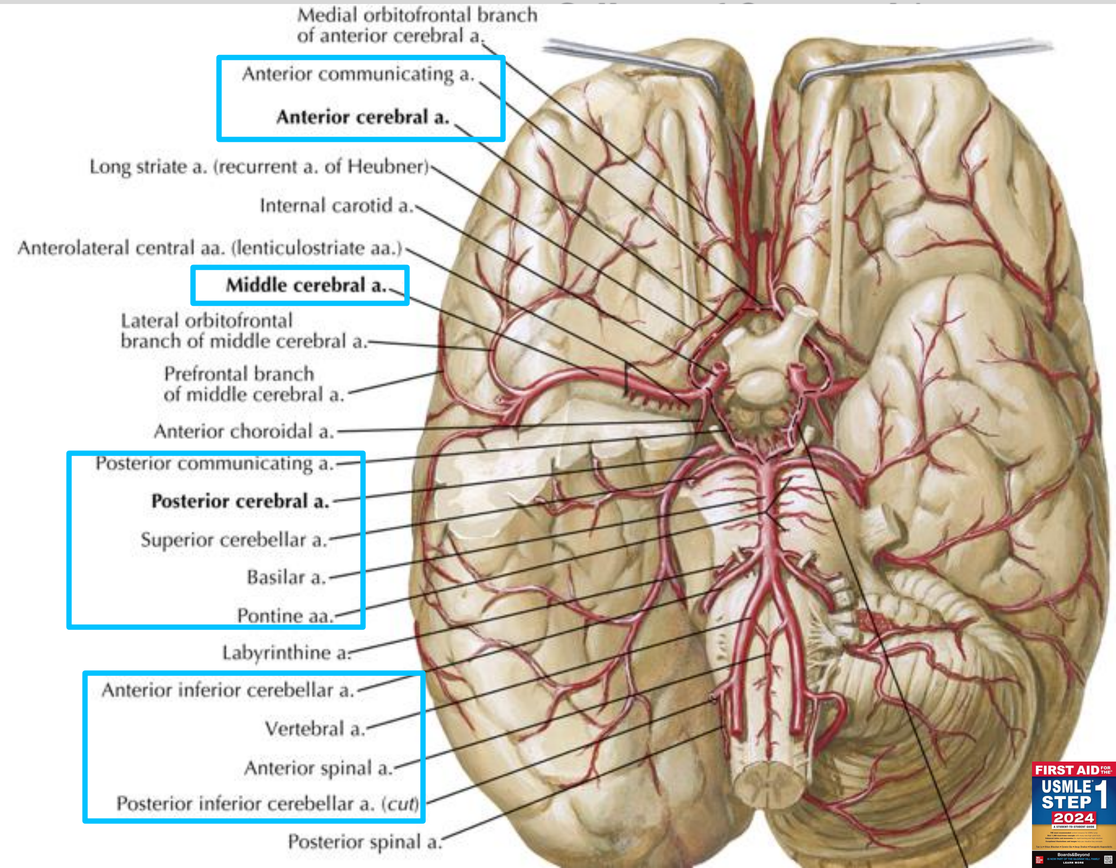
- Vertebral artery →**1 Anterior spinal artery (ASA)**
  - Supply: anterior spinal cord (mostly motor)
- Posterior inferior cerebellar artery (PICA) → **2 Posterior spinal artery (PSA)**
  - Supply: posterior spinal cord (mostly sensory)



# Circle of Willis (CoW)

NEW YORK INSTITUTE OF TECHNOLOGY

- A circular network of arteries at the **base of the brain** that provides **collateral circulation**.
- Formed by **the joining of the internal carotid arteries** (anterior circulation) and the vertebral arteries (posterior circulation) via **PCom**.
- **Acts as a safeguard**: if one major artery is blocked, the Circle of Willis can potentially provide alternate routes for blood flow, reducing the risk of ischemia.
- **Common site for the development of saccular (berry) aneurysms**, particularly at branch points such as the **ACom**.
- **Anatomical variations** are common in the Circle of Willis, with complete and functional circles present in only about 30-50% of individuals.

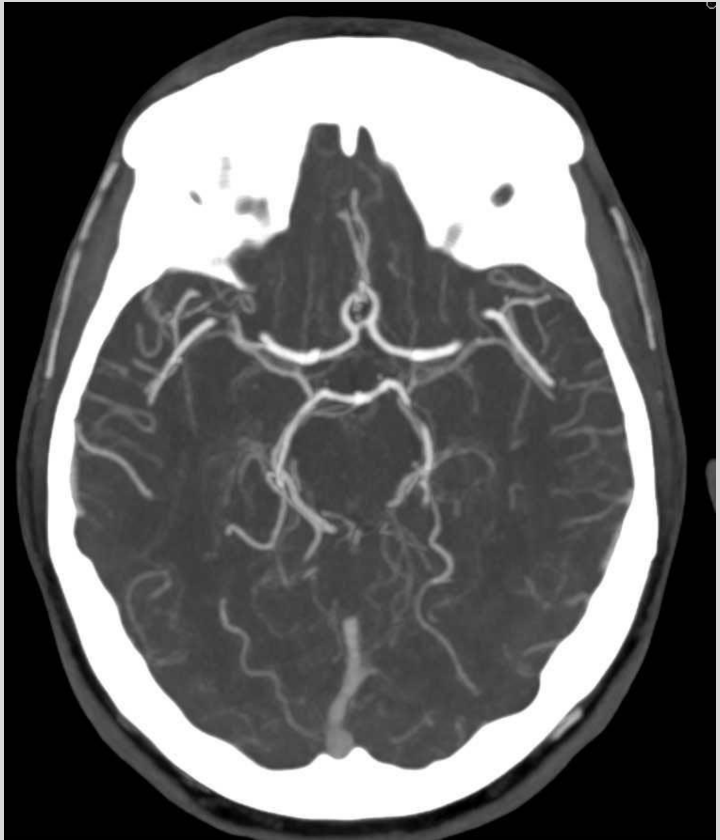


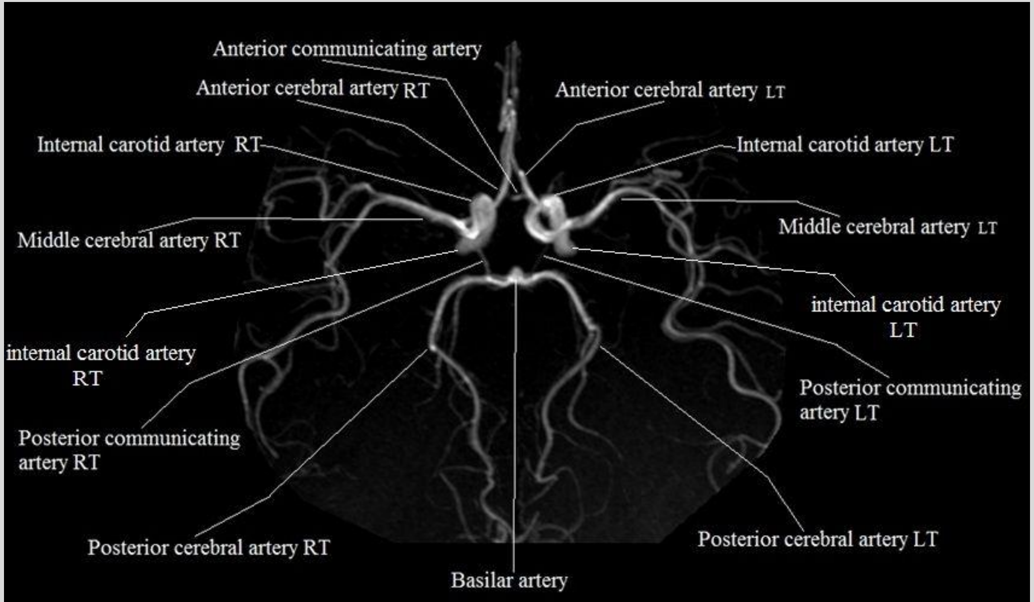
FIRST AID FOR THE USMLE STEP 1 2024

# Circle of Willis- Can you label the CTA (CT angiography)?

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine



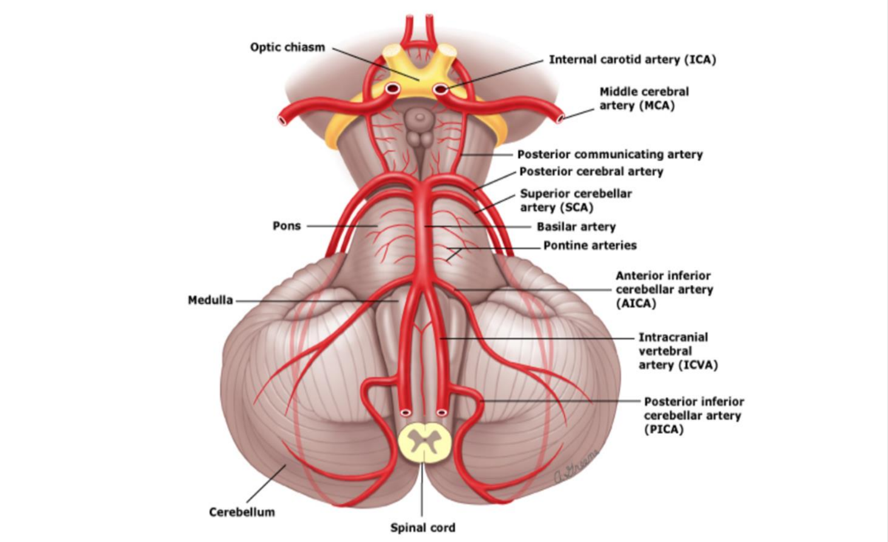


NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

# Knowledge check:

# POSTERIOR CIRCULATION



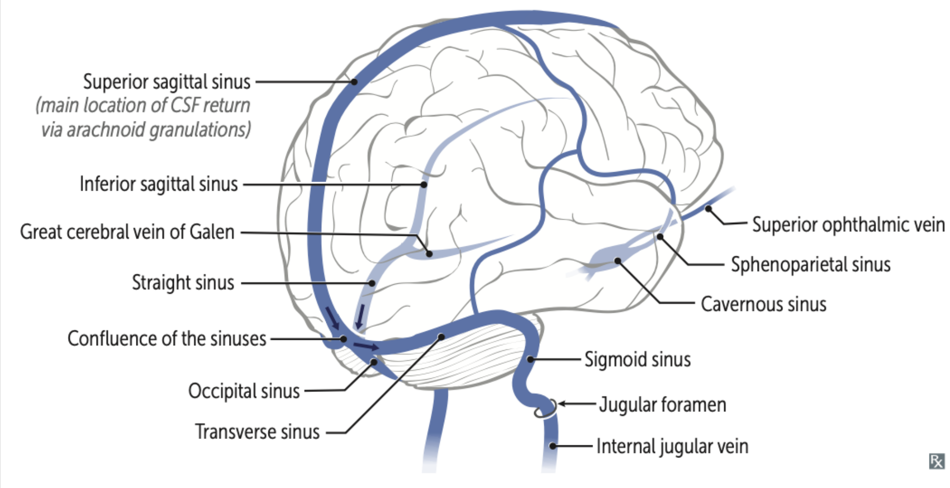

1. **Parent vessel**→ where does it come from?
2. **Destination** → is it giving another branch or supplying a brain area?
3. **Adjacent structures** → what other neurovascular structures are relative to it?
4. **Blood supply** →what part of the brain does it supply blood to?

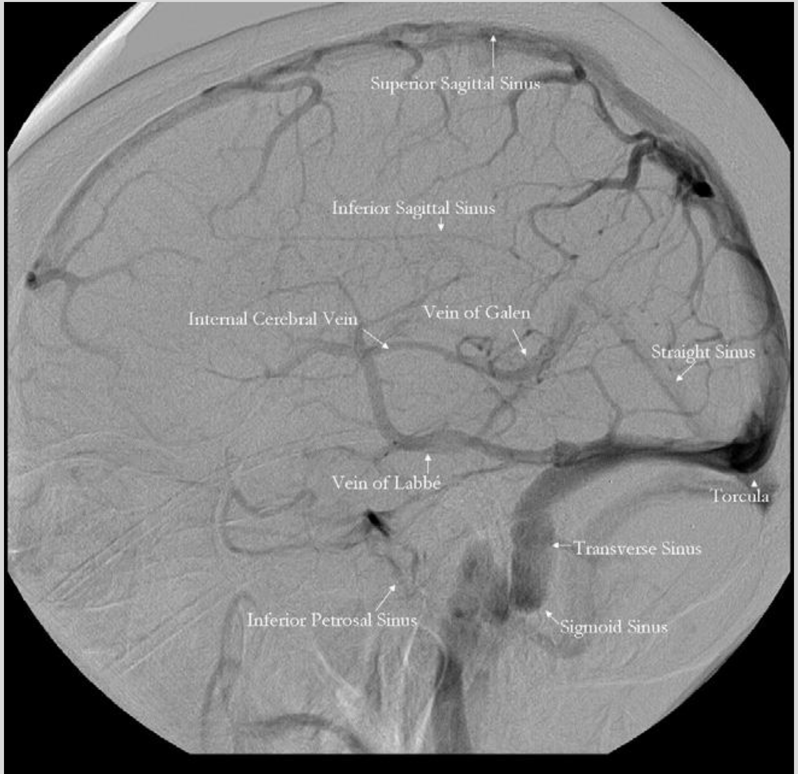
# CEREBRAL VENOUS SYSTEM

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Cerebral sinuses
- Deep cerebral veins
- Superficial cerebral veins

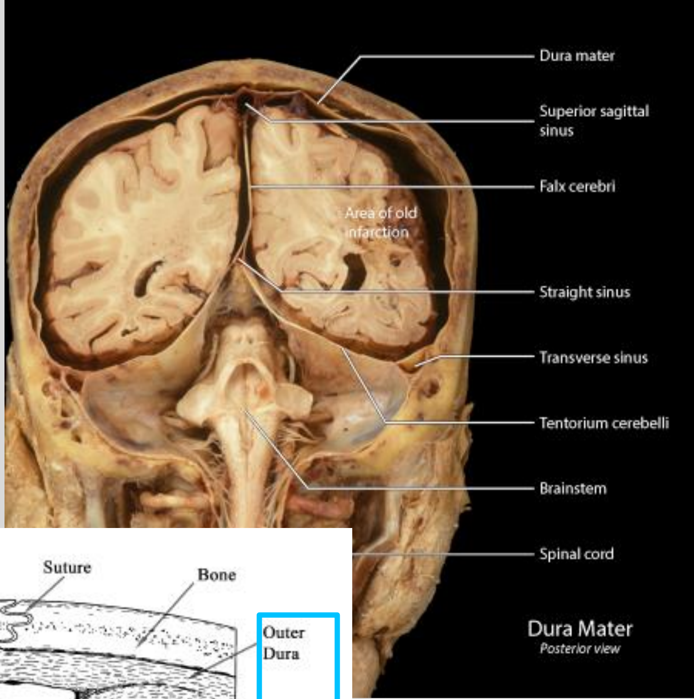


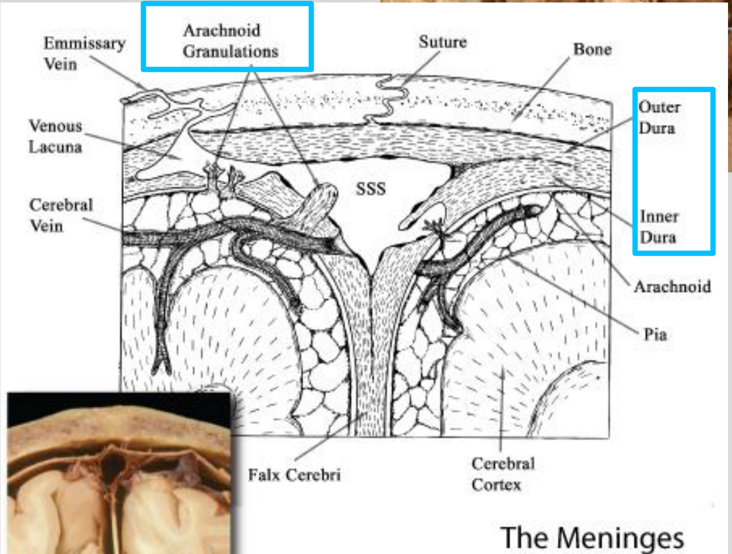



# Cerebral Sinuses

Appropriately called **dural sinuses**.

- **Channels between layers of dura mater** that **drain deoxygenated blood** and **cerebrospinal fluid** (CSF) from the brain.
- While referred to as venous structures, they are **not veins**
  - Lack the 3 tunica layers
  - Lack valves
  - Walls are simply dural mater
- Made by the **infoldings** of **falx cerebri** and **tentorium cerebelli**
- Have small **outpouching** on their surface called **arachnoid granulations**:
  - Drain CSF from subarachnoid space to venous system



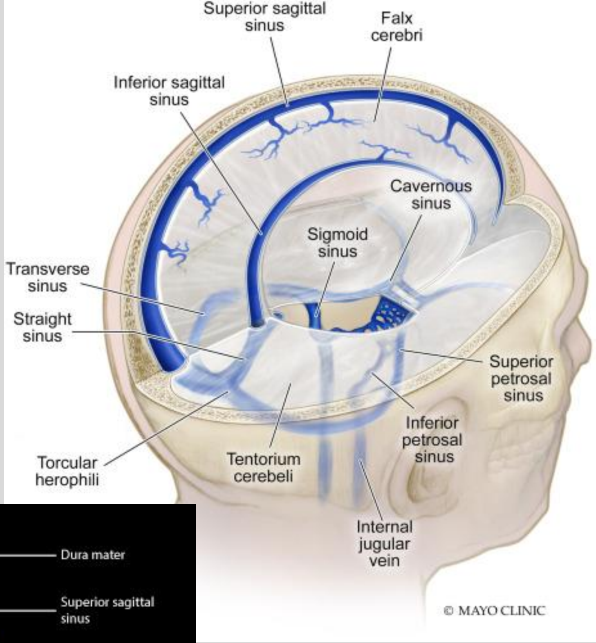



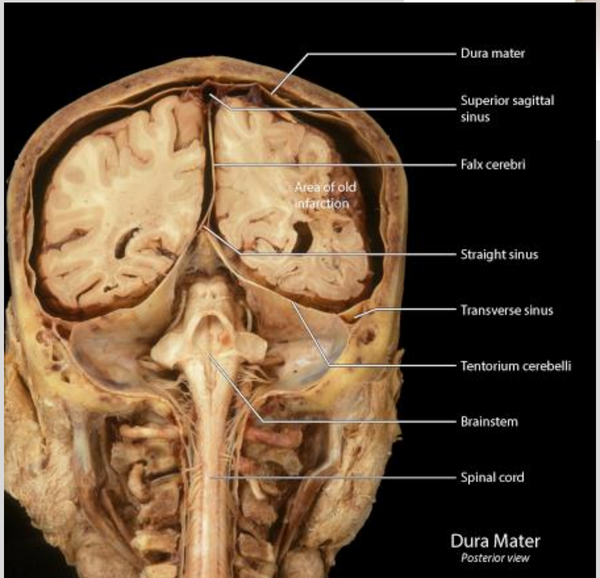

The Meninges

# Cerebral Sinuses

**Major dural sinuses:**

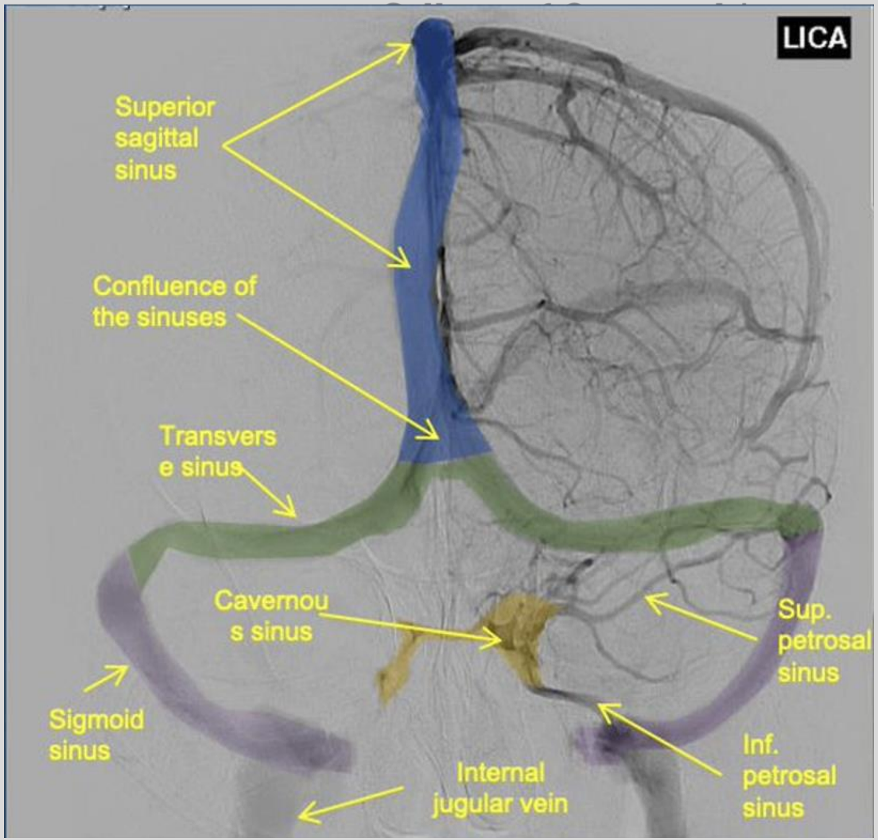
- **Superior sagittal sinus**: Drains blood from the superficial veins and flows into the confluence of sinuses.
- **Inferior sagittal sinus**: Joins the straight sinus, draining deep brain structures.
- **Straight sinus:** Connects the inferior sagittal sinus and the great vein of Galen.
- **Cavernous sinus**: unique dural venous structure drains orbit and face.
- **Transverse sinuses**: Receive blood from the confluence of sinuses and drain into the sigmoid sinuses.
- **Sigmoid sinuses**: Drain into the internal jugular veins (IJVs), completing the venous outflow.
- **Confluence of sinuses**/ Torcular herophili : Meeting point of superior sagittal, straight, and transverse sinuses.



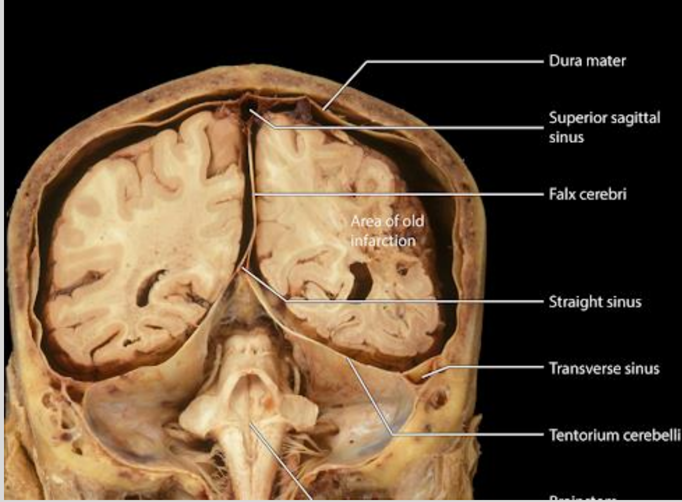



# Dural Venous Sinuses

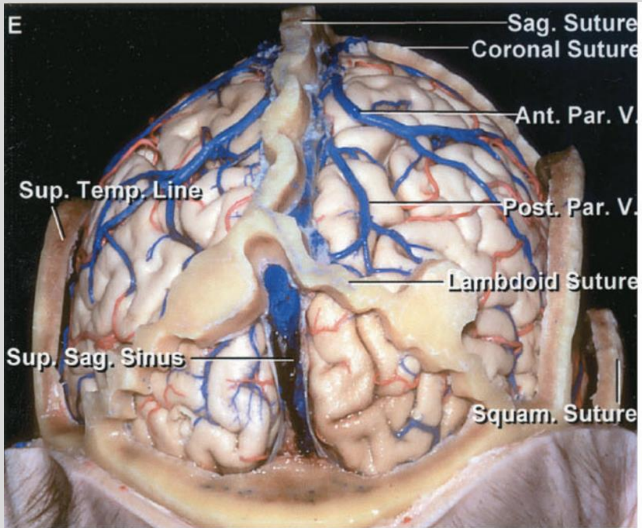
NEW YORK INSTITUTE OF TECHNOLOGY

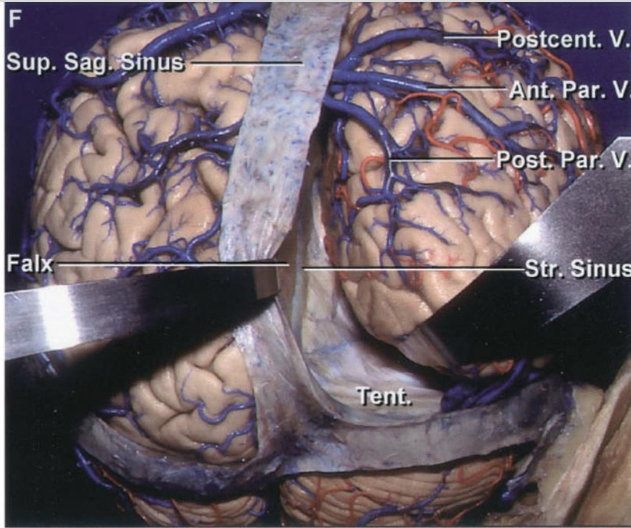



# Superior Sagittal Sinus (SSS)




- Formed by the **two layers of dura that makes up the falx cerebri; periosteal dura and dura propria** (meningeal dura).
- Travels **superior and medial over the cerebral cortex** right **below the sagittal suture**.
- It **joins the confluence of sinuses posteriorly** (torcula)
- The **SSS drain blood** from the **superficial cerebral veins** overlying the cortex called **bridging veins**.
- These veins can be injured during trauma to the head and cause **subdural hematoma (chronic venous bleed)**



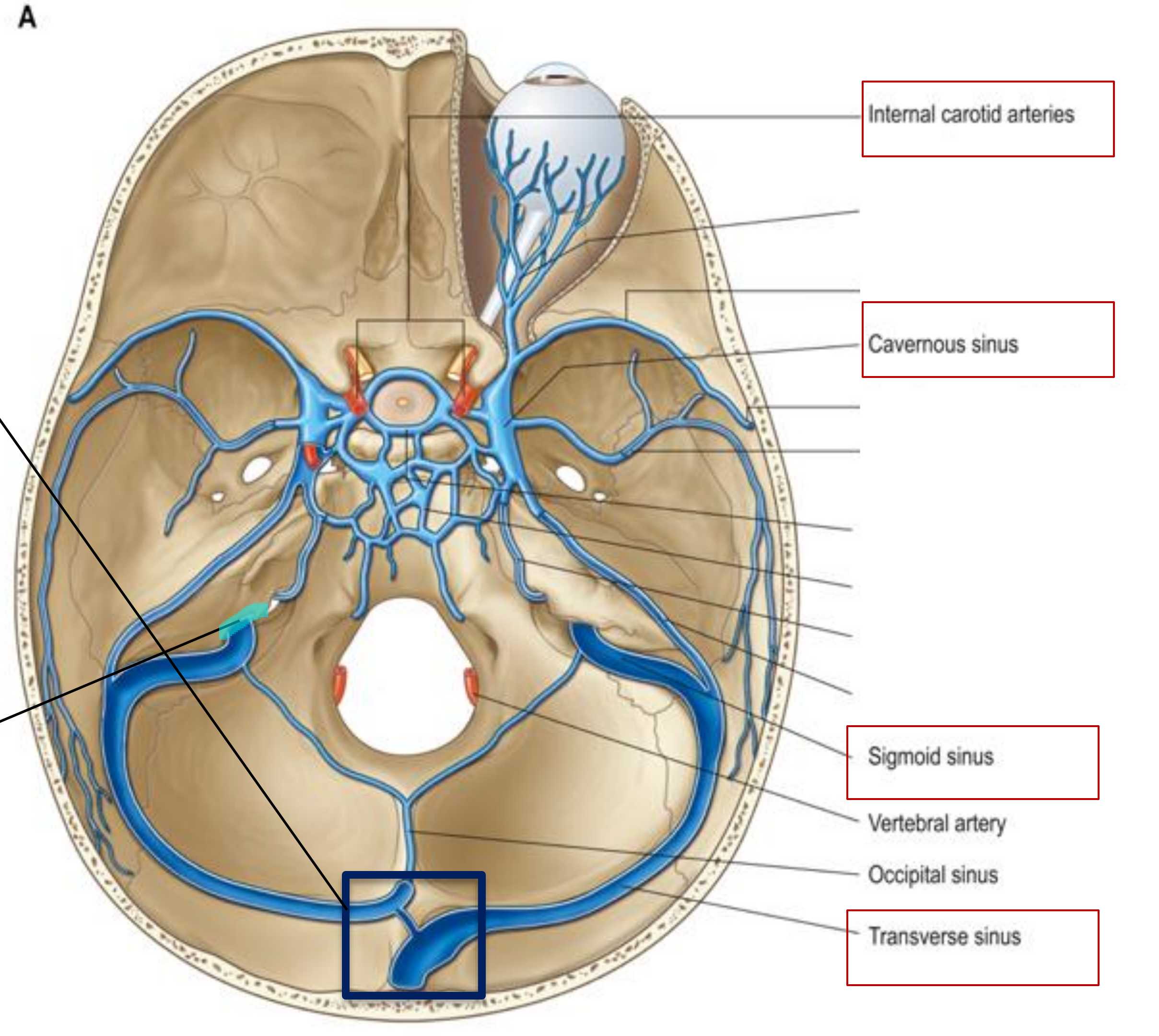
# Cerebral Sinuses

## Transverse Sinus

- Originates at the **confluence of sinuses in the posterior** skull, lies **laterally in the groove of occipital bone**.
- **Drains blood from the superior sagittal sinus, straight sinus, and occipital sinus.**

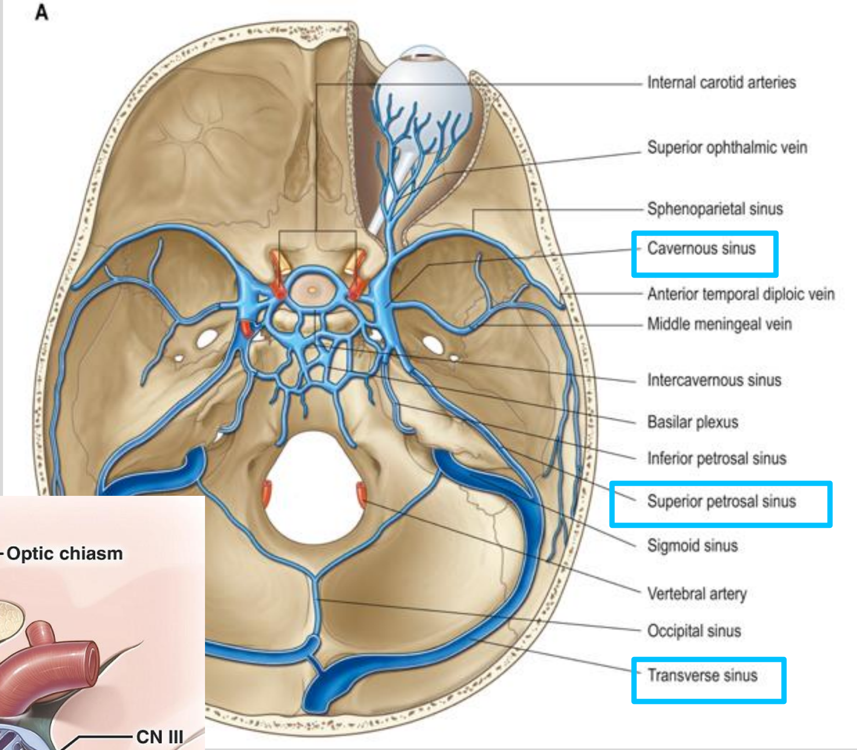
## Sigmoid Sinus

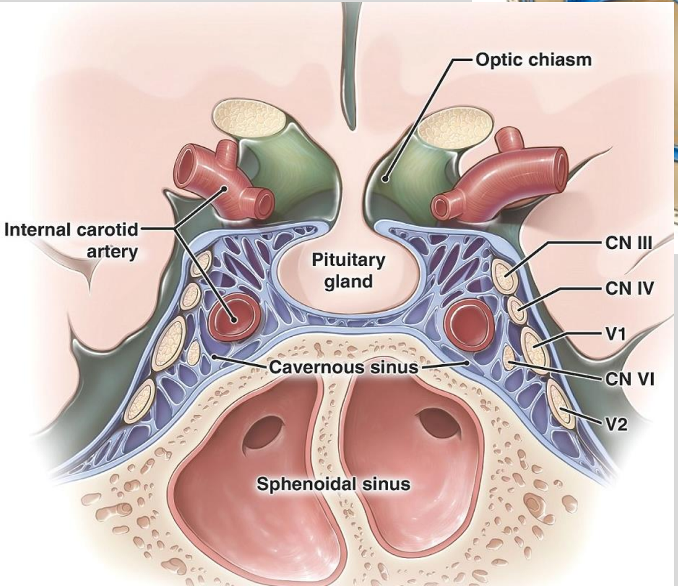
- Continuation of the transverse sinus at the '**transverse-sigmoid junction**"
- Courses through the skull base to join the **internal jugular vein** at the **jugular foramen**.
- **Proximity to cranial nerves and the jugular foramen**; important in neurosurgical approaches.



# Cavernous sinus

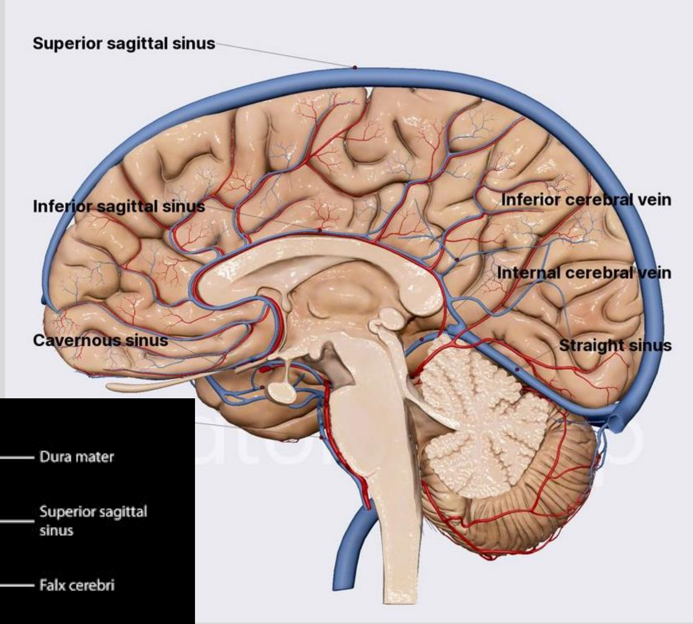
- Located on **either side of the Sella turcica**, important for **draining the orbit and face**.
- **Contains cranial nerves** (III, IV, V1, V2, VI) and the **internal carotid artery**.
- **Lesions of cavernous portion of ICA** (aneurysms, cavernomas etc.) generally cause **compression of CN VI** due to its medial proximity.
- Drains blood from superficial cerebral vein into transverse sinus via superficial petrosal vein.

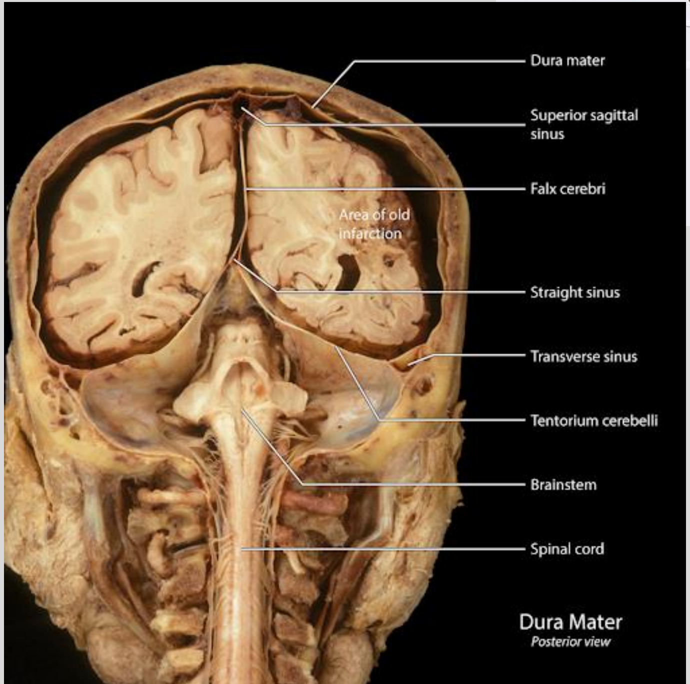


# Deep cerebral veins

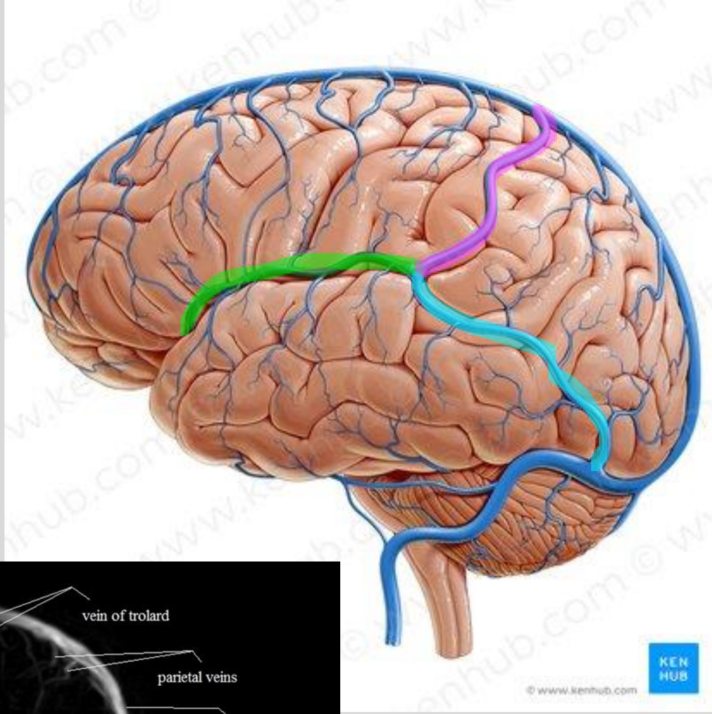
- **Drains deeper structures** such as the basal ganglia, thalamus, and internal capsule
- Major veins: Great cerebral vein of Galen, internal cerebral veins, basal veins of Rosenthal- **don't memorize**
- **Deep cerebral veins → straight sinus → confluence of sinuses → transverse sinus → sigmoid sinus → internal jugular vein**
- Straight sinus is found at the junction between the **falx cerebri** and the **tentorium cerebelli**

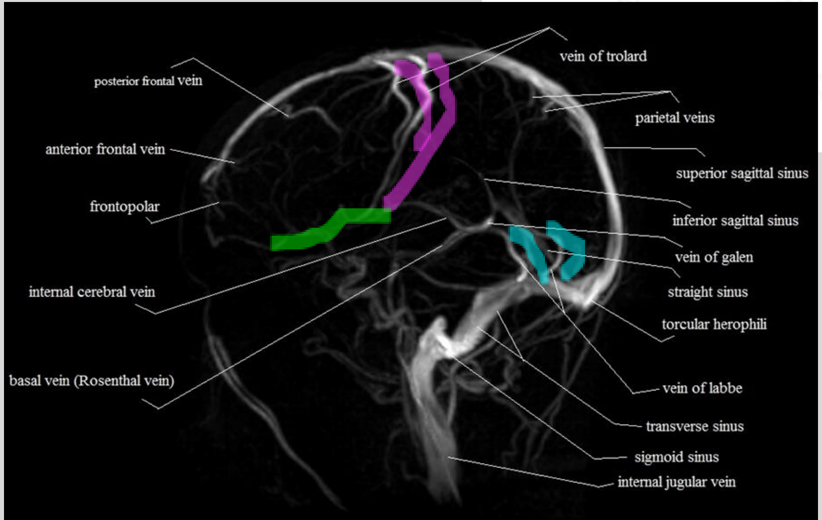


# Superficial cerebral veins

- Primarily drains the cerebral cortex.
- Major veins:
  - **Superficial middle cerebral vein** or Sylvian vein
  - **Vein of Trolard** or Superior anastomotic cerebral vein
  - **Vein of Labbe** or Inferior anastomotic cerebral vein



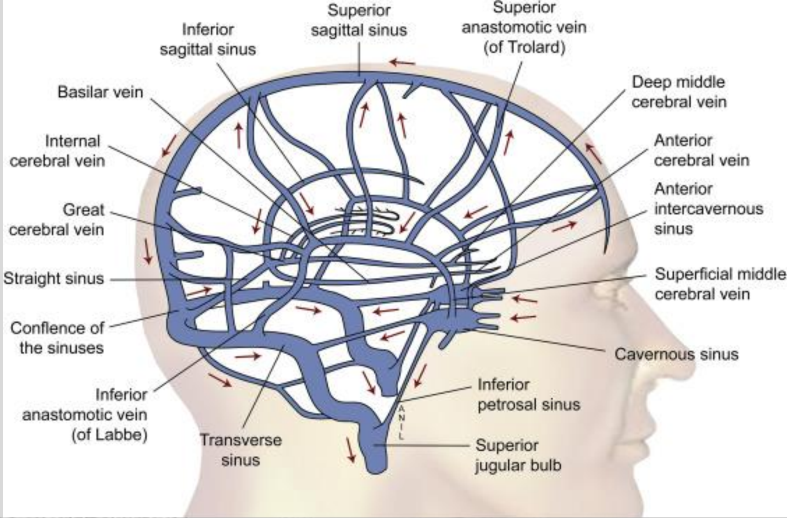
# Direction of Drainage

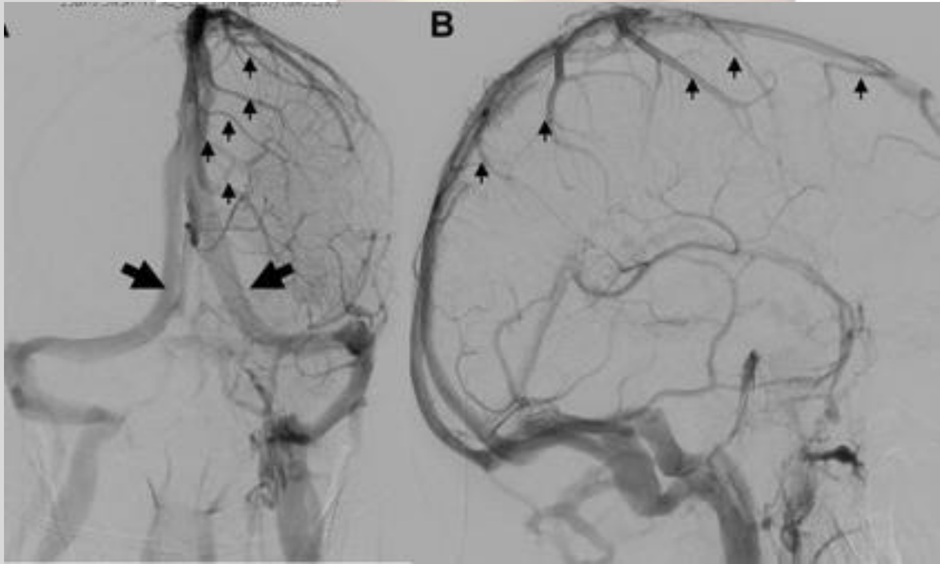
**Superficial Venous System:** Drains the cerebral cortex into the **superior sagittal sinus and transverse sinuses**

**Deep Venous System:** Drains deep brain structures into the **internal cerebral veins and straight sinus**

Blood from the superficial and deep venous systems converges at the **confluence of sinuses**

From here, it drains into the **transverse sinuses**, then to the **sigmoid sinuses**, and exits via the **internal jugular veins** through the **jugular foramen**



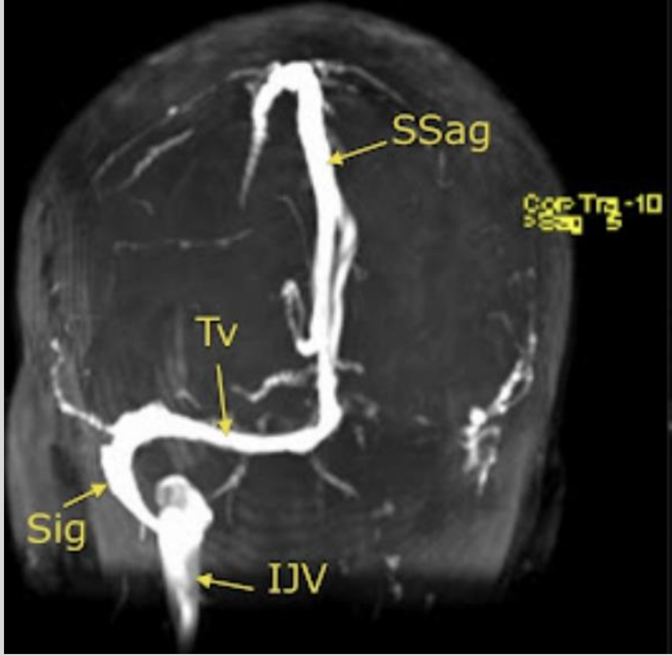
# Cerebral Venous Thrombosis (CVT)

**CVT is a blood clot in the dural venous sinuses or cerebral veins**, **blocking venous drainage** from the brain. Common sites:

- **Superior sagittal sinus**
- **Transverse sinus**
- **Sigmoid sinus**
- **Cortical veins**

**Risk Factors: Hypercoagulable states** (e.g., **pregnancy, oral contraceptive use, malignancy**), infections (e.g., mastoiditis, sinusitis), head injury, chronic inflammatory diseases (e.g., lupus, IBD)

**Clinical Presentation:** Headache (often severe and sudden), seizures, focal neurological deficits (e.g., weakness, vision changes), symptoms of increased intracranial pressure (e.g., papilledema, nausea, vomiting).




MRV: Venous thrombosis of left transverse sinus at the junction of confluence of sinuses

More on this in the cerebrovascular pathology lecture later..

# Knowledge check:

# CEREBRAL VENOUS SYSTEM

NEW YORK INSTITUTE OF TECHNOLOGY

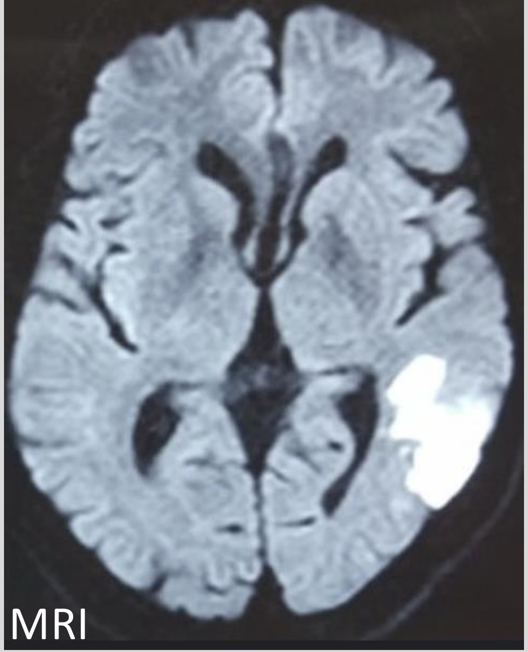
College of Osteopathic Medicine

1. **Parent Vessel→** Which major veins or sinuses does this vessel drain to?
   a. Is it part of the superficial or deep venous system?
2. **Destination→** Where does this venous pathway lead (e.g., confluence of sinuses, internal jugular vein)?
   a. Does it converge with other veins or directly exit the cranial vault?
3. **Adjacent Structures→** Which neurovascular structures are located near this sinus or vein?
   a. Are there any dural folds, cranial nerves, or arterial branches in close proximity?
4. **Drainage Territory→** What regions of the brain does this sinus or vein drain (e.g., cortex, deep brain structures)?

# Clinical question

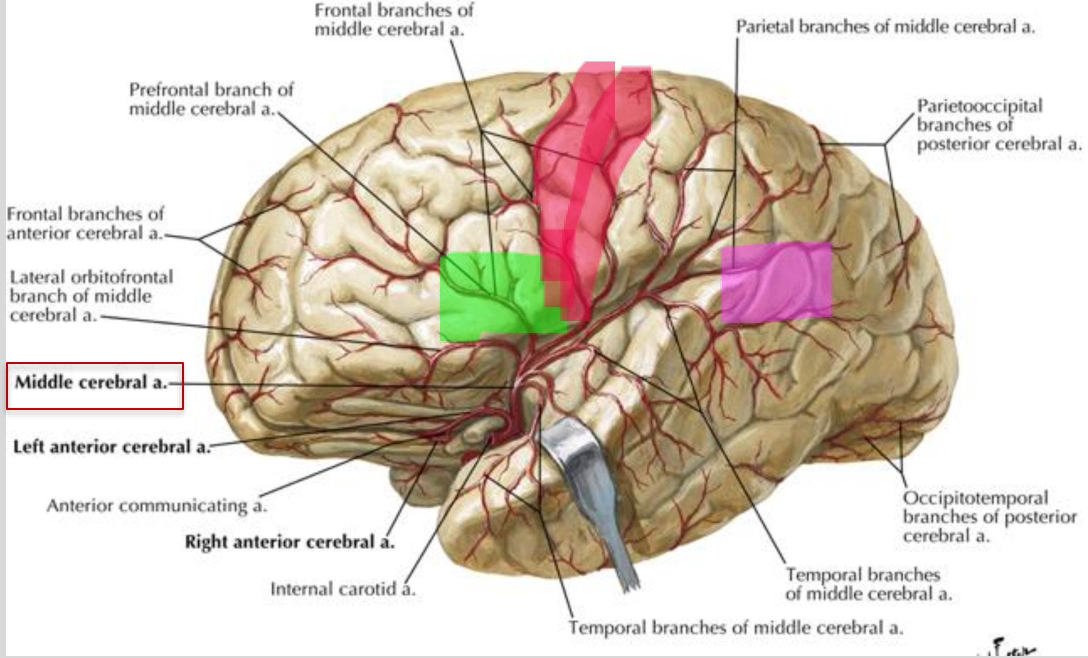
A 65-year-old man is brought to the emergency department with sudden difficulty understanding language. He speaks fluently but uses nonsensical words and does not comprehend spoken or written language. MRI shows an ischemic stroke affecting a specific artery. Which artery is most likely involved in causing these symptoms?

**A.** Anterior cerebral artery
**B.** Middle cerebral artery
**C.** Posterior cerebral artery
**D.** Basilar artery
**E.** Vertebral artery

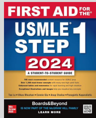


# Broca’s and Wernicke’s Area supply

- **Broca's area**, in the opercular and triangular parts of the inferior frontal gyrus.
    - Lesion here results in **expressive aphasia (fluency)**
- **Left precentral gyrus**
    - **primary motor cortex**
- **Wernicke’s area**, in the posterior superior temporal gyrus.
    - Lesion here results in **receptive aphasia (understating)**

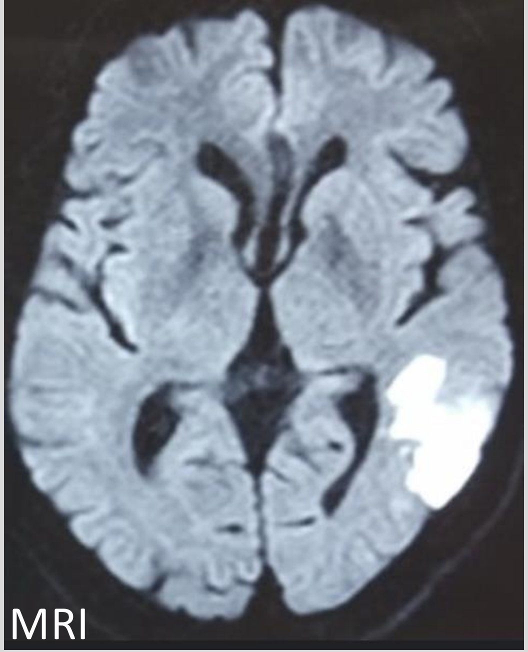



**MCA Territory**



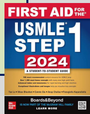
# Clinical question

A 65-year-old man is brought to the emergency department with sudden difficulty **understanding language**. **He speaks fluently but uses nonsensical words** and does not comprehend spoken or written language. MRI shows an **ischemic stroke affecting a specific artery**. Which artery is most likely involved in causing these symptoms?




**A.** Anterior cerebral artery
**B. Middle cerebral artery**
**C.** Posterior cerebral artery
**D.** Basilar artery
**E.** Vertebral artery

The **middle cerebral artery (MCA)** supplies **the lateral aspects of the cerebral cortex**, including **Wernicke's area** in the dominant hemisphere. An infarct in this region can lead to **Wernicke's aphasia**, characterized by **fluent speech without meaningful content and poor language comprehension**. This is commonly associated with a **distal MCA stroke.**



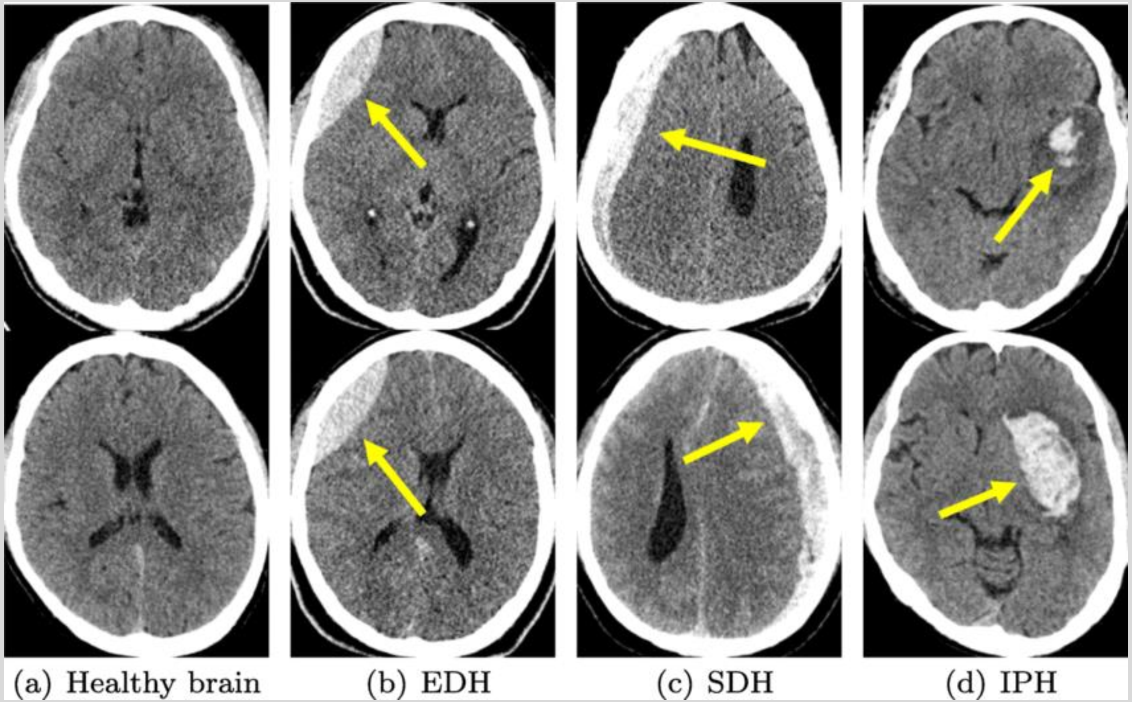
# Intracranial Bleeds- brief overview

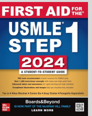
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- Epidural hematoma
- Subdural hematoma
- Intraparenchymal hematoma
- Intraventricular hematoma
- Subarachnoid hemorrhage

1. Location of the bleed
2. Cause
3. Presentation
4. Imaging



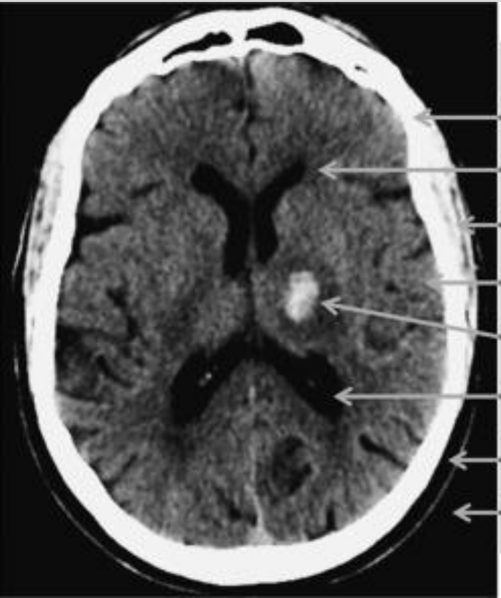

More on this in the cerebrovascular pathology lecture later..

# HEAD CT (w/o contrast) Quick Reference

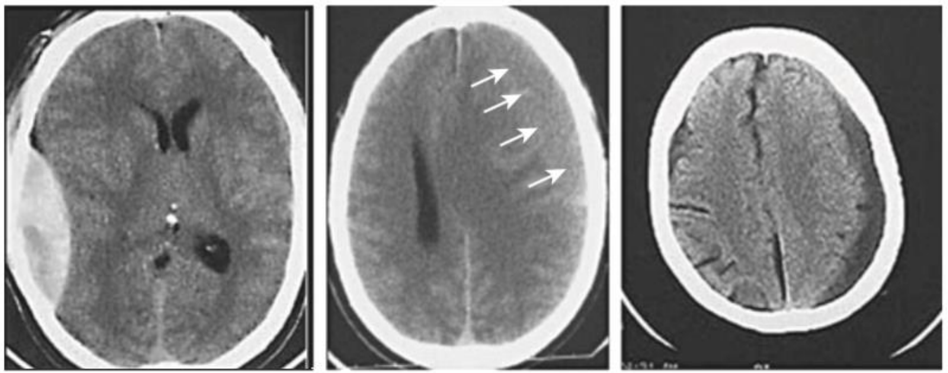
## YOU ARE NOT TESTED ON THIS

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic



| CT Number Ranges in Hounsfield Units (HU) | |
|---|---|
| Bone | +1000 |
| White matter | +20 to 30 |
| Muscle | +20 to 40 |
| Gray matter | +30 to 40 |
| Hemorrhage | +65 to +95 |
| CSF (water) | 0 |
| Fat | -30 to -70 |
| Air | -1000 |



| Acute | Subacute | Chronic |
|---|---|---|
| **HYPER**DENSE | **ISO**DENSE | **HYPO**DENSE |

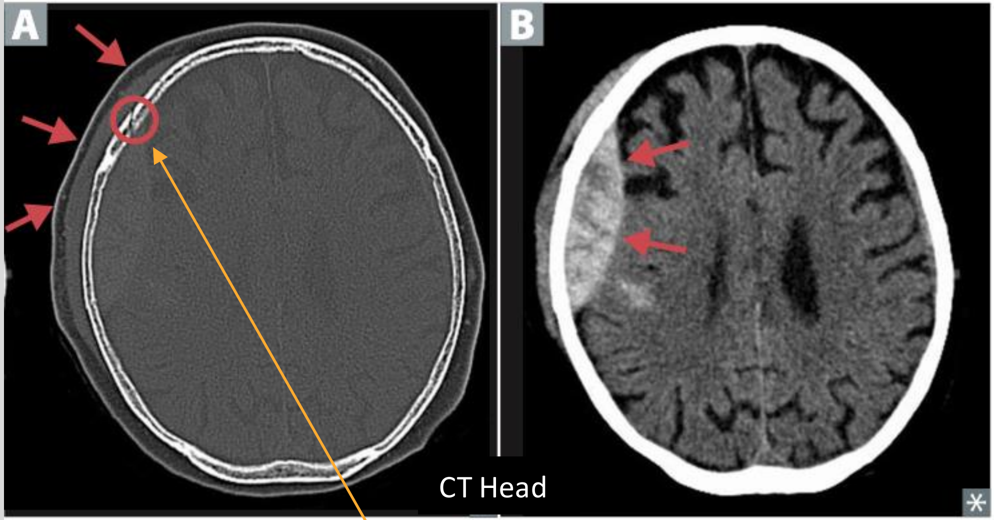
# Epidural Hematoma

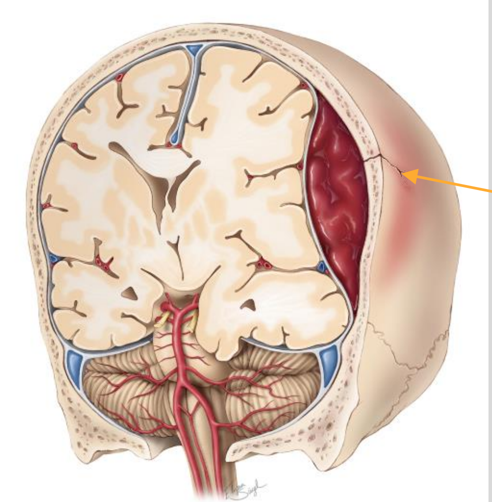
**Location:** **Between the skull and the periosteal layer of dura**

**Cause:** Often due to trauma, particularly with skull fractures that damage the **middle meningeal artery**

**Presentation:** "Lucid interval" followed by rapid deterioration; symptoms include headache, vomiting, and potential loss of consciousness

**Imaging:** **Lens-shaped** (biconvex) on CT scan



The Neurosurgical Atlas

Almost always associated with **skull fracture**

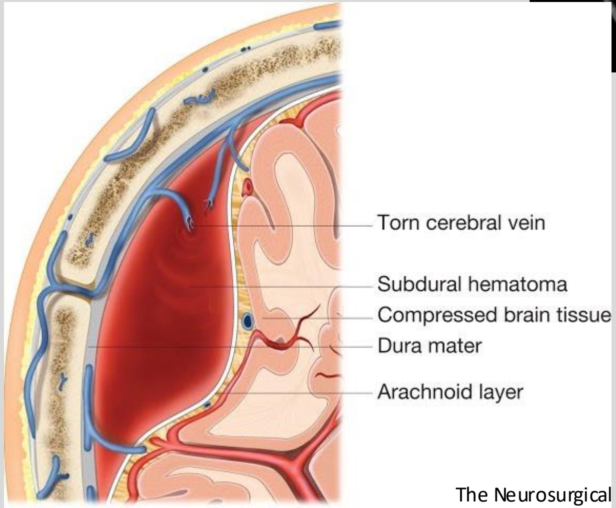
# Subdural Hematoma


**Location: Between the dura and arachnoid mater**

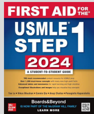
**Cause:** Usually from tearing of **bridging veins**, often in the elderly or in those with brain atrophy; associated with minor trauma or falls→ **chronic subdural**

**Presentation:** Symptoms may develop over hours to weeks; includes headache, confusion, and focal neurological deficits

**Imaging: Crescent-shaped** (concave) bleed on CT scan




The Neurosurgical Atlas



# Subarachnoid Hemorrhage (SAH)

**Location:** In the **subarachnoid space, where cerebrospinal fluid (CSF) circulates**

**Cause:** Most commonly due to **trauma**, followed by **ruptured aneurysms** (e.g., **berry aneurysms**)

**Presentation:** Sudden onset of a severe "**thunderclap" headache ("worst headache of my life**"), neck stiffness, photophobia

**Imaging: Blood in the subarachnoid space on CT**


Normal head CT


CT Head

Classic starburst pattern of hemorrhage: Ruptured aneurysmal SAH

# Intraparenchymal Hematoma

NEW YORK INSTITUTE OF TECHNOLOGY

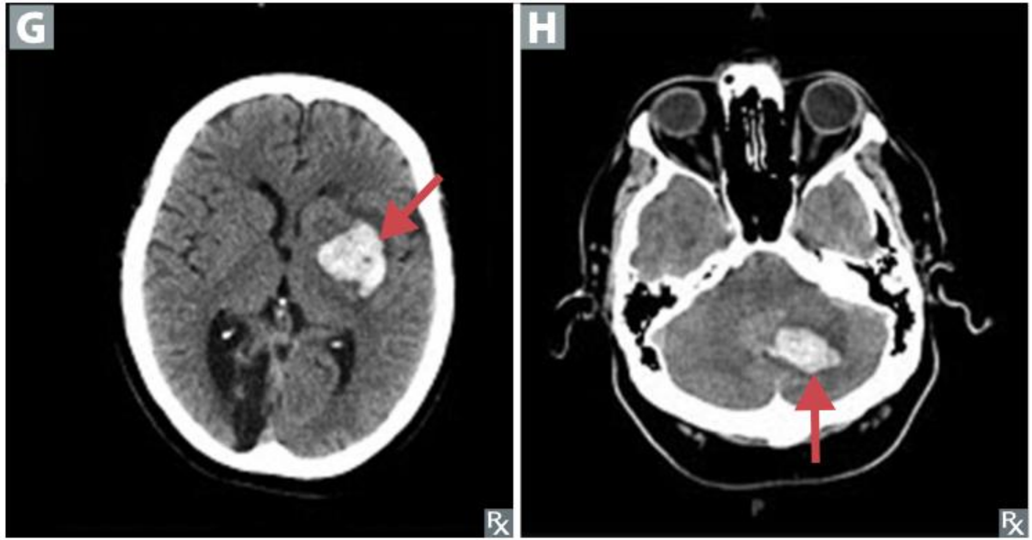
College of Osteopathic

**Location:** Within the **brain parenchyma** itself

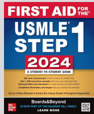
**Cause:** Often due to **hypertension, cerebral amyloid angiopathy**, anticoagulation therapy, or trauma

**Presentation:** Focal neurological deficits depending on the hemorrhage location, often with sudden onset

**Imaging:** **Hyperdense area on CT**, usually within the brain tissue




CT Head



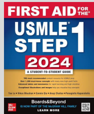
# Intraventricular Hematoma

**Location:** Within the **brain's ventricular system**.

**Cause:** Can result from **extension of intracerebral hemorrhage** or trauma, commonly seen in **preterm infants**.

**Presentation:** Variable; may cause symptoms of **hydrocephalus** if blood obstructs CSF flow.

**Imaging:** Blood in the ventricular spaces on CT.

# Clinical question

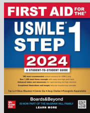
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

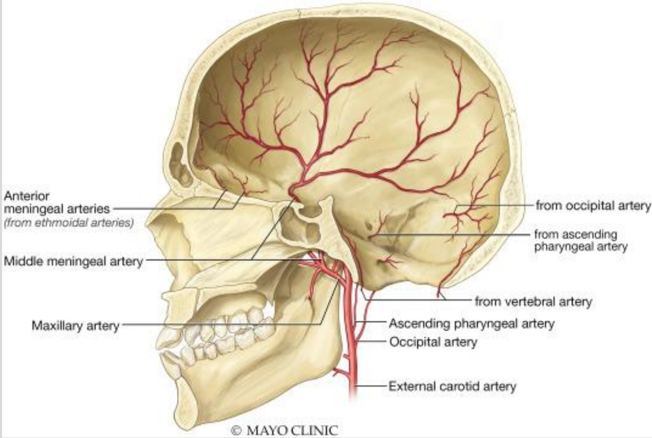
A 28-year-old man presents to the emergency department following a motor vehicle accident. He initially lost consciousness, then experienced a lucid interval but is now worsening with severe headache, nausea, vomiting, confusion, and one-sided weakness. His vital signs show hypertension and bradycardia, and he begins to have episodes of altered consciousness. A CT scan reveals a lens-shaped hyperdensity.

Which vessel is most likely ruptured in this type of bleed?

**A.** Middle cerebral artery
**B.** Anterior cerebral artery
**C.** Bridging dural veins
**D.** Middle meningeal artery
**E.** Inferior anastomotic cerebral vein



# Clinical question

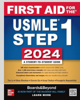



A 28-year-old man presents to the emergency department following a **motor vehicle accident**. He initially **lost consciousness**, then experienced **a lucid interval** but is now worsening with severe headache, nausea, vomiting, confusion, and one-sided weakness. His vital signs show hypertension and bradycardia, and he begins to have episodes of altered consciousness. A CT scan reveals **a lens-shaped hyperdensity**.

Which vessel is most likely ruptured in this type of bleed?

**A.** Middle cerebral artery
**B.** Anterior cerebral artery
**C.** Bridging dural veins
**D. Middle meningeal artery**
**E.** Inferior anastomotic cerebral vein

An epidural hematoma typically results from rupture of the **middle meningeal artery** following trauma, often with an **associated skull fracture**. This leads to a lens-shaped (biconvex) appearance on CT and is commonly characterized by a lucid interval followed by rapid neurological deterioration.



# Summary

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

**Arterial Supply:**

**Anterior Circulation**

Supplied by the internal carotid arteries, branching into the anterior and middle cerebral arteries.

Key areas supplied: frontal lobes, parietal lobes, and superior temporal lobes.

**Posterior Circulation**

Supplied by the vertebral arteries, which form the basilar artery and branch into the posterior cerebral arteries.

Key areas supplied: brainstem, cerebellum, occipital lobes, and inferior temporal lobes.

**Venous Drainage:**

**Superficial System**: Includes the superior sagittal sinus and cortical veins draining into dural venous sinuses.

**Deep System**: Drains deep structures of the brain via the internal cerebral veins into the straight sinus and then to the confluence of sinuses.

# Summary

NEW YORK INSTITUTE OF TECHNOLOGY

...opathic

**Watershed Areas:**

- Border zones between major arteries, particularly vulnerable to ischemia during low blood flow states.

**Key Clinical Correlates:**

- **Cerebral Venous Thrombosis (CVT):** Obstruction in the dural sinuses or cerebral veins, leading to increased intracranial pressure, venous congestion, and potential infarction.
- **Stroke Syndromes**: Result from blockages in specific arteries; symptoms depend on the artery affected.
- **Intracranial Bleeds**: Epidural, subdural, subarachnoid, and intraparenchymal hemorrhages, each with distinct causes, locations, and imaging features.

# Thank you!

Please email me with any questions:

vmathe03@nyit.edu

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

# Lecture and Lab Feedback Form:

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R